When Does Training on Downscaled Images Yield the Same Gradients?

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Abstract

Diffusion transformers deliver strong image generation, but their training cost grows superlinearly with resolution. Recent work justifies training or sampling at reduced resolution on a spectral premise: at high noise, a downscaled latent preserves almost the full surviving signal. Whether a downscaled step also preserves the native *training gradient* signal, however, has remained unresolved. Measured at the gradient level, what downscaling changes turns out to be the residual of a near-cancellation: a data-side and a compute-graph-side perturbation, each far larger than the realized gap, strongly anti-aligned (pooled $\rho \approx -0.91$) at every measured route and noise level — measured in full on one adapter, and replicated across the mid-noise window on two further adapters of the same base model. On that residual an effective two-term account holds: a noise-dependent term governed by the downscale *ratio*, which decays at high noise as the spectral premise predicts, and a σ-independent floor governed by the target grid's *absolute token count*, removed by no noise level. The measured (route, σ) map corroborates the account and uncovers structure the spectral picture cannot express but the cancellation geometry anticipates: on the 1024→768 route, a window ($0.65 < \sigma < 0.95$), predicted by no spectral criterion at any tolerance, where the downscaled gradient stays within a small margin of the native one. Training LoRA adapters with downscaled steps on the measured-safe 1024→896 route above a one-sided noise gate ($\sigma > 0.5$) reduces training time by 14.6% at a fixed step budget while remaining near-native in weight space. Code is available at https://github.com/sorryhyun/anima_lora.

1 Introduction

The cost of a forward pass in a diffusion transformer (Peebles & Xie, 2023), the backbone of current text-to-image systems (Esser et al., 2024; Chen et al., 2024), grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. This cost structure makes the title question worth stating precisely: can the *gradient* of an adapter be computed on a downscaled latent as a direct substitute for the native one, with the model, the objective, and the σ distribution unchanged and only the spatial grid changed on some steps?

There is a well-studied reason to expect an affirmative answer *at high noise*. As the noise level σ grows, per-frequency signal-to-noise falls, and high spatial frequencies are the first to be masked by the noise, so that above a spectral crossover a downscaled latent carries *almost* the full surviving signal. Studies of this coupling, however, are concentrated on the *inference* side: scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), and spectrally-guided noise schedules (Esteves & Makadia, 2026) run high-noise *sampling* steps at reduced resolution and judge the premise by the generated output. Where resolution is reduced during training (Jin et al., 2024; Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023), the low-resolution stage is given its own, modified objective, and success is again judged from the final samples. By the spectral premise, substitution is safe *whenever the noise has masked the frequencies the coarse grid loses*.

This evidence, however, leaves the question open at *training* time: whether a downscaled step delivers the native gradient direction under an unchanged objective. Output-level success observes the substitution only through the confounds of a full training or sampling pipeline, and a small difference in a quality score does not imply a small difference in what a training step *learns*. What a training step consumes is the expected adapter gradient, and substituting a downscaled, re-encoded latent for the native one at training time perturbs more than the network's input: the regression target carries the clean image at unit weight at every noise level, and the compute graph itself changes with the grid. We therefore decompose the perturbation at its source, into a data branch and a graph branch — and, measured, the two branches are individually large and strongly anti-aligned, so that the gap between the downscaled and native gradients is the residual of a near-cancellation. On that residual we state an effective two-term law, a noise-dependent term set by the downscale *ratio* and a σ-independent term set by the target grid's *token count*, and measure each directly.

The contributions of this paper are as follows:

1. **Two accounts of the gap, and a debiased measurement that scores them** (§3, §4). We define the substitution gap at the gradient level and state two accounts of it in the same units. The first is the gap implied by the existing spectral argument, taken in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026); it depends on the downscale *ratio* alone by construction. The second, the *gradient-perturbation* account, treats the substitution as a perturbation of both factors of $g = J^{\top} r$. Measured, the two perturbation branches nearly cancel — strongly anti-aligned at every route and noise level, an anti-alignment that is uniform across depth and module type and invariant to the adapter operating point on the measured checkpoint, and that replicates on two further adapters of the same base model on the 768 route — and on the residual of that cancellation the account reduces to a route amplitude on a universal mismatch curve *plus* a σ-independent term set by the target grid's token count. The two differ structurally before any measurement (§3.2), and the resulting (route, σ) map (§4.2), read with each route's finite-sample bias estimated and subtracted and against a pre-specified margin, decides between them: it uncovers on the 1024→768 route a window ($0.65 < \sigma < 0.95$) that the spectral account does not predict at any tolerance. Scored head to head on the same curves, the spectral account errs at 0.147–0.355 RMSE, whereas ours predicts held-out routes at ~0.07–0.09, and the token-count law correctly predicts the σ-independent term on both held-out routes.
2. **Selective low-resolution training** (§5). Restricting the substitution to the measured-safe 1024→896 route above a one-sided gate ($\sigma > 0.5$) gives an opt-in σ-conditional trainer that reduces the forward-compute footprint by 15.1% and training time by 14.6% at a fixed step budget, with a weight-space endpoint within kernel-noise reach of a native retrain, rising to endpoint cosine 0.75 when the substitution is scheduled late at proportionally reduced saving (Table 1), the low-excess 1024→768 window stacking onto the late rule at no endpoint cost; a visual counterpart (Fig. 5) confirms the endpoint renders stay close to native.

2 Related work

Flow matching and DiT. Flow matching (Lipman et al., 2023) regresses a velocity field along prescribed probability paths between noise and data; rectified flow (Liu et al., 2023) takes the paths to be straight lines, giving the linear noising $z_\sigma = (1-\sigma)x + \sigma\epsilon$ and the constant target $v = \epsilon - x$ used throughout, the training objective of current large text-to-image systems (Esser et al., 2024). The Diffusion Transformer (Peebles & Xie, 2023) replaces the U-Net denoiser with a transformer over patchified latent tokens, so a spatial grid enters the network only as a token sequence: token count grows quadratically with edge length and attention cost quadratically again in token count, the cost structure that makes resolution the expensive axis. Spatial position is now widely carried by a grid-dependent coordinate system, most commonly rotary embeddings (Su et al., 2024).

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogeboom et al., 2023) train at increasing resolution as a schedule, each low-resolution stage carrying its own stage-specific objective. A more recent family ties resolution to the noise level itself, on the spectral premise above, and has so far been studied on the *inference* side. Scale-wise distillation (SwD) (Starodubcev et al., 2025) states the claim directly (above the crossover, the noised downscaled latent is close in distribution to the downscaled noised latent) and samples early steps at reduced resolution. Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives the argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model, a criterion on the per-band output of the *Bayes-optimal predictor* rather than merely on input SNR, from which an inference-time resolution schedule follows. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same reasoning to schedule design; where the premise does enter training, as in pyramidal flow matching (Jin et al., 2024), the objective is restaged per pyramid level rather than left native.

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* that matches the downscaled grid's relative phase geometry to native, in order to decompose the resolution-sensitivity floor, and adopt the σ-gated band schedule of Zhao et al. (2026) as a training-time refinement. On the inference side, Wu et al. (2025b) measure a content-free rotary attention bias and show position-interpolated rescaling samples its sinusoidal phase filter at the wrong phases, ranking interpolation worst among extension schemes at the sample level; their setting is mixed-resolution attention within one forward pass, whereas our substitution is a whole-grid change at training time (no mixed-scale attention arises) and our attribution is at the training-gradient level.

Noise levels as tasks. A gradient-level literature treats the timesteps of diffusion training as tasks and measures their pairwise interference: task affinity, the cosine between per-timestep loss gradients, decays smoothly as the noise-level gap grows (Go et al., 2023) and turns negative between distant timesteps (Ma et al., 2025), motivating interval clustering and decouple-then-merge training. Those studies compare full gradients *across* noise levels; our object is different — the product-rule split of one gradient's *change* under input downscaling at a fixed noise level — and the near-cancellation central here has no counterpart in that setting. We cite them as the precedent for smooth, σ-indexed gradient-direction structure in these models.

3 Two accounts of what demotion costs

This section fixes the estimand (§3.1), then states two accounts of it in the same units: a training-gap interpretation built on the spectral account of the inference-side literature (§3.2), and the gradient-perturbation account (§§3.3–3.4). The two are not variants of one model: they disagree about which quantity governs a route's cost, about what happens at pure noise, and about whether a single parameter can order all routes.

3.1 The estimand

Objective and noising. A flow-matching model regresses the constant-velocity transport between a clean latent and pure noise. With clean latent x, noise $\epsilon \sim \mathcal{N}(0, I)$, and noise level $\sigma \in (0, 1]$, the noised input is $z_\sigma = (1-\sigma)\,x + \sigma\,\epsilon$ and the target is the velocity $v = \epsilon - x$; with caption condition c the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$.

Demotion. We write $e_0 \!\to\! e$ for a *demotion*: the native latent at the route's source edge e_0 is replaced with a coarser-grid one at its target edge e. With y the image at its native resolution, $\mathcal{E}$ the VAE encoder, and $\mathcal{R}_e$ the pixel-space downscale to the tier-e bucket (native aspect ratio preserved), $x_{\mathrm{src}} = \mathcal{E}(y) \longrightarrow x_{\mathrm{dem}} = \mathcal{E}\big(\mathcal{R}_e(y)\big)$, yielding a latent with proportionally fewer spatial tokens. The demoted step trains on $z_\sigma = (1-\sigma)\,x_{\mathrm{dem}} + \sigma\,\epsilon$ with target $v = \epsilon - x_{\mathrm{dem}}$, so the substitution changes both what the network sees and what it is asked to predict, but only through the spatial grid.

The estimand. For an arm $a \in \{\mathrm{src}, \mathrm{reenc}, \mathrm{dem}\}$, where the source arm *src* trains on the native-resolution latent x_{src}, the demoted arm *dem* trains on x_{dem}, and the control arm *reenc* trains on the native latent decoded and re-encoded at native resolution, let $\bar{g}_a(\sigma) = \mathbb{E}_\epsilon\,[\nabla_\theta \mathcal{L}]$ be the population per-bin mean adapter gradient of a query (image, caption). The *population demotion distance* of the route, $d_{e_0 \to e}$, is

$$d_{e_0\to e}(\sigma) \;=\; 1 - \cos\big(\bar{g}_{\mathrm{src}}(\sigma),\ \bar{g}_{\mathrm{dem}}(\sigma)\big) \;=\; \tfrac{1}{2}\,\big\|\hat{g}_{\mathrm{src}} - \hat{g}_{\mathrm{dem}}\big\|^2, \tag{1}$$

with $\hat{g}$ the unit-normalized gradients. Route-level scalars carry the *route* as a subscript, abbreviated to the target edge alone (d_e) wherever the source is fixed by context; the control's distance d_{reenc}

carries its arm label instead, since it changes no grid (full convention: Appendix A.1). The *reported* quantity of every bin-resolved map and every fit in this paper is the *re-encoding excess*

$$\mathrm{gap}_e(\sigma) \;:=\; d_e(\sigma) \;-\; d_{\mathrm{reenc}}(\sigma), \tag{2}$$

The control changes no grid, so its distance d_{reenc} prices the VAE round trip alone; subtracting it isolates the part of d_e attributable to the grid change, and, unlike d_e, the excess can be negative. Every map in the body aggregates per example, taking the cosine per image and then the mean; the batch-aggregate object that SGD follows in batched training is a distinct estimand, treated in Appendix A.1.

3.2 The spectral account

The spectral argument of §2 supplies the field's working model of the same gap; we now restate it in the units of Eq. 1. Its strongest available form is the tolerance-parameterized crossover of Xiao et al. (2026).

Demotion deletes a band. Write $u^{(\omega)}$ for the frequency-ω component of a field u in the latent's spatial Fourier decomposition, with ω measured in cycles per sample of the *native* grid, so that the native Nyquist frequency sits at $\frac{1}{2}$. The downscale $\mathcal{R}_e$ resamples the image with e/e_0 times fewer samples per axis, and a grid at that density can represent no frequency above half its own sampling rate; the resampler's anti-aliasing prefilter removes, rather than aliases, whatever lies above. Setting aside the VAE round trip, which the reenc control of §3.1 is constructed to absorb, demotion is an ideal low-pass:

$$x^{(\omega)}_{\mathrm{dem}} \;=\; \begin{cases} x^{(\omega)}, & |\omega| < \omega_e, \\ 0, & |\omega| \geq \omega_e, \end{cases} \qquad \omega_e \;=\; \frac{1}{2}\,\frac{e}{e_0}\,, \tag{3}$$

mapping every band at or above the coarse grid's Nyquist frequency ω_e to zero and acting as the identity on every band below it.

Carried to the training gradient. The noising $z_\sigma = (1-\sigma)\,x + \sigma\,\epsilon$ is linear and hence diagonal in the spatial Fourier basis, so it acts on each band independently. Modeling each clean-latent band as independently Gaussian, $x^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ with P_ω the band's clean-latent power, Xiao et al. (2026) attach to each band an activation time: a crossover t_ω past which the Bayes-optimal per-band velocity prediction agrees, to within a tolerance δ, with its data-independent limit $\epsilon^{(\omega)}$. The destroyed bands of Eq. 3 all sit at or above ω_e; on a decaying spectrum the lowest of them is the most powerful, so it crosses last, and the safe-substitution boundary for the whole route sits at t_{ω_e}. The account attributes the route's whole cost to those bands: whatever they contribute to the residual, and through the backward pass to the adapter gradient, perturbs $\bar{g}_{\mathrm{dem}}$ away from $\bar{g}_{\mathrm{src}}$. The implied gradient gap is that contribution, gated at the Nyquist band's activation time:

$$\mathrm{gap}^{\mathrm{spec}}_e(\sigma) \;=\; S_e(\sigma)\,\mathbf{1}\big[\sigma < t_{\omega_e}\big], \qquad t_\omega \;=\; \frac{1}{1 + \sqrt{\delta \,/\, \big(P_\omega\,(1 + P_\omega - \delta)\big)}}, \tag{4}$$

where $S_e(\sigma)$ is the destroyed-band contribution to the distance of Eq. 1, and the tolerance δ is the account's single free parameter, instantiated in Appendix E. At the residual level the premise is complete enough to compute S_e: the Gaussian posterior mean is the classical per-band Wiener shrinkage (Wiener, 1949), so the cross-grid mean-residual mismatch is confined to the destroyed band, ordered across routes by destroyed-band energy, and follows in closed form from the power spectrum (Appendix A.3). Carrying it into the gradient units of Eq. 1 costs one calibrated gain, since the map passes through the network's Jacobian, about which a spectral model of the *data* says nothing. Appendix E instantiates this pipeline when the account is scored.

Whatever the estimate of S_e, the account above commits to three limits, fixed by the *form* of Eq. 4 before any gradient is measured. **(a) One parameter orders every route**: all boundaries move together under the single tolerance, so the family is totally ordered. **(b) Ratio only**: the cut $\omega_e = \frac{1}{2}\,e/e_0$ of Eq. 3 is a function of the downscale *ratio* alone, so the estimate is structurally blind to any absolute-size effect. **(c) No bias term**: for every route and every δ the predicted gap vanishes above a finite boundary, so no σ-independent contribution is expressible.

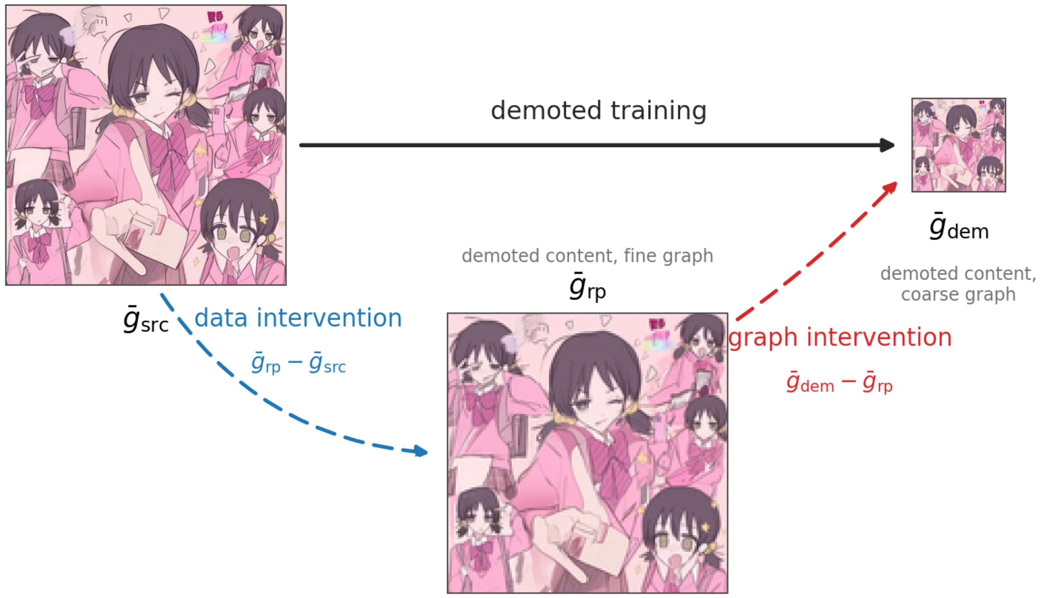

Figure 1: **The three arms of Eq. 5.** *src* runs the ordinary training step on the native latent $\mathcal{E}(y)$ (fine graph). The *repromote* arm rp trains on the same image demoted and then carried back to the native resolution in pixel space, $\mathcal{E}(\mathrm{upscale}(\mathcal{R}_e(y)))$ — demoted content on the native grid — so src $\to$ rp changes only the data the objective consumes. *dem* trains on the demoted latent $\mathcal{E}(\mathcal{R}_e(y))$ on its own coarse grid, so rp $\to$ dem changes only the compute graph. The two dashed steps compose exactly to demoted training itself (the solid arrow): rp is the demotion path's hidden waypoint — training never realizes it, but it is the one realizable intermediate that splits the path into its two interventions. The pipeline cost shared by the two demoted inputs is priced by the reenc control of §3.1, against which the measured data leg is re-referenced (§4.3). Content: the model's own render (prompt, Appendix G); for visibility the downscale is drawn at ratio $\frac{1}{3}$ and the repromote upscale nearest-neighbor.

3.3 The gradient-perturbation account: what demotion perturbs

We start one level lower, from what the substitution touches in the gradient itself. Differentiating the loss of §3.1 gives, per draw and up to a constant,

$$g \;=\; \nabla_\theta \tfrac{1}{2}\big\|\hat{v}_\theta - v\big\|^2 \;=\; J^\top r, \qquad r \;=\; \hat{v}_\theta - v, \qquad J \;=\; \partial \hat{v}_\theta / \partial\theta,$$

with r the prediction residual and J the Jacobian through which the compute graph turns a residual into an adapter gradient. Demotion perturbs both factors: the residual, through the data the objective consumes, and the Jacobian, through the compute graph. The estimand is built from noise means, so split each factor into its noise mean and fluctuation under the averaging that defines $\bar{g}_a$ in §3.1, $r_a = \bar{r}_a + \tilde{r}_a$ and $J_a = \bar{J}_a + \tilde{J}_a$; the arm's mean gradient is then

$$\bar{g}_a(\sigma) \;=\; \mathbb{E}_\epsilon\big[J_a^\top r_a\big] \;=\; \bar{J}_a^\top \bar{r}_a(\sigma) \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_a^\top \tilde{r}_a\big], \qquad \bar{r}_a(\sigma) \;=\; \mathbb{E}_\epsilon\big[\hat{v}_\theta - v\big],$$

and the *mean residual* $\bar{r}_a$ is a velocity-shaped field with a definite σ-curve for each arm.

The two factors are perturbed by two physically distinct interventions — the data the objective consumes, and the compute graph that pulls it back — and the pair is realizable one intervention at a time: a *repromote* arm rp, the demoted input carried back to the native grid, applies the data intervention while holding the graph fixed (Fig. 1). The branches are the two partial differences along this path,

$$\delta g \;=\; \bar{g}_{\rm dem} - \bar{g}_{\rm src} \;=\; \underbrace{\big(\bar{g}_{\rm rp} - \bar{g}_{\rm src}\big)}_{\text{data branch } B_e} \;+\; \underbrace{\big(\bar{g}_{\rm dem} - \bar{g}_{\rm rp}\big)}_{\text{graph branch } C_e}, \tag{5}$$

a telescoping identity in the shared adapter-parameter space where all three gradients live: exact at every amplitude, with nothing discarded — the terms a mean-factor product rule would collect as a remainder (the coupling of the two perturbations, and the cross-arm differences of the noise-covariance couplings $\mathbb{E}_\epsilon[\tilde{J}_a^\top \tilde{r}_a]$) ride *inside* the legs, assigned to whichever intervention realizes them along the path (Appendix A.2). The assignment is unique because the path is: the fourth corner of the data–graph lattice, native content on the coarse grid, does not exist (Fig. 7), so no alternative split is realizable — and no cross-grid matrix product is ever computed: J_{src} and J_{dem} act on different grids, and only the legs themselves are measured. One mean-factor object does recur downstream: writing $\bar{r}_{\text{dem}} = \bar{r} + \Delta\bar{r}(\sigma)$ (bare bars denote the source arm), the *cross-grid mean-residual mismatch* $\Delta\bar{r}(\sigma)$ is the same data intervention the data branch reads in gradient space, read at the residual level — and, unlike any per-branch matrix product, its norm is directly measurable (§4.4). Read against Eq. 5, the spectral account of §3.2 keeps only the input-mediated part of the data branch, gated at Nyquist, and sets the rest to zero.

Measured, the decomposition is not a large term plus a small correction. At every measured (route, σ) of the verdict campaign both branches are individually far larger than the realized gap, and their components orthogonal to $\bar{g}_{\text{src}}$ are strongly anti-aligned: writing $\rho = \cos(B_e^\perp, C_e^\perp)$, the vector-resolved probe of §4.3 reads $\rho \in [-0.96, -0.63]$ at every non-degenerate (route, σ) cell, pooling to a single constant $\bar{\rho} = -0.91$ over the gated verdict bins — a pooling §4.3 licenses by measurement rather than assumption: the anti-alignment is route-uniform, uniform across depth and module type, and invariant to the adapter operating point. The measured gap curve is therefore the *residual of a near-cancellation*: the data branch is what demotion does to the data the objective consumes, the graph branch is what the changed compute graph does in response, and the two slide against each other along nearly one line, so the estimand retains only what the cancellation misses. That anticipates structure the branch magnitudes alone cannot produce: wherever the amplitude ratio $|B_e^\perp|/|C_e^\perp|$ passes through 1 while the anti-alignment stays deep, the residual collapses — below either branch alone, and below the route's own floor. The mid-σ cliffs of the measured map sit at this ratio's passage through 1, and the 768 route's in-window dip where the ratio comes closest to balance (§4.2, §4.3).

Two interventions follow immediately from the decomposition, rather than from experimental convention, and each isolates a branch. At $\sigma = 1$ the input is pure noise in every arm, carrying no trace of the image, so the input-mediated part of the data branch vanishes by construction; the regression target, however, carries the clean image at unit weight at every σ, so the mean residual retains an image-dependent survivor, $\bar{r}_a(1) = x_a - \mathbb{E}[x \mid c]$ up to the frozen model's approximation error (Appendix A.3): the endpoint reads the graph branch plus that target-mediated remnant of the data branch. Zeroing the image in input *and* target removes the survivor as well: the arms then differ in nothing but the grid, every data-mediated term vanishes by construction, and the x-zero probe reads the graph branch alone. The one question the two probes cannot settle, whether the survivor is *resolvable* in the angular estimand, is discharged in §4.4.

3.4 The angular expansion and the two-term reduction

It remains to read Eq. 5 in the units of the estimand. Split the perturbation δg into a rescaling of $\bar{g}_{\text{src}}$ and a rotation away from it, with $\kappa_\parallel$ and $\kappa_\perp$ the two components in units of $\|\bar{g}_{\text{src}}\|$. The gap is then an exact saturating function of a single rotation-to-scale ratio,

$$d_e \;=\; 1 - \frac{1}{\sqrt{1+\kappa_{\text{eff}}^2}}\,, \qquad \kappa_{\text{eff}} \;=\; \frac{\kappa_\perp}{1+\kappa_\parallel}\,, \tag{6}$$

so only the orthogonal component moves the estimand, and the cosine is *blind* to a parallel rescaling. Expanding at small perturbation and substituting Eq. 5 gives the *four-term angular expansion*: writing $u^\perp$ for the component of u orthogonal to $\bar{g}_{\text{src}}$,

$$d_e(\sigma) \;\approx\; \underbrace{\frac{\|B_e^\perp\|^2}{2\,\|\bar{g}_{\text{src}}\|^2}}_{\text{data share } S_e} \;+\; \underbrace{\frac{\|C_e^\perp\|^2}{2\,\|\bar{g}_{\text{src}}\|^2}}_{\text{graph share } \Phi_e} \;+\; \underbrace{\frac{\langle B_e^\perp,\, C_e^\perp\rangle}{\|\bar{g}_{\text{src}}\|^2}}_{\text{projected interaction } I_e} \;+\; R'_e, \tag{7}$$

with both shares non-negative and an interaction that can carry either sign. The legs being exact (Eq. 5), the remainder R'_e is *purely geometric*: it contains only the cosine expansion's own beyond-quadratic terms, no branch-level content; the derivations and its expanded form are in Appendix A.4.

From the expansion to a measurable form. On the reported excess, Eq. 7 is exact but not yet an instrument: no term in it is separately measurable per bin. The excess itself does part of the work: the reenc arm changes no grid, so its graph leg vanishes identically, its expansion degenerates to $d_{\text{reenc}} \approx S_{\text{reenc}} + R'_{\text{reenc}}$, and subtracting it cancels only the re-encode component of the data share, passing the graph share and interaction through untouched. The rest is carried by one measured term, one domain rule, and two assumptions, each assumption tested by a designated probe in §4:

- **(i) Geometric remainder, handled by domain**: with the legs exact, R'_e carries only the expansion's higher-order terms, which grow with perturbation amplitude. It enters as a reading rule rather than a smallness assumption: wherever the measured amplitudes leave the quadratic domain, magnitudes are read through the un-expanded exact link (Eq. 6) — the unit convention stated below, under which nothing is truncated.
- **(ii) Interference as measured**: the projected interaction I_e is *not* assumed small — it is the dominant structure, large and negative at every in-window condition (the near-cancellation of §3.3; numbers in §4.3). It enters the reduction as a measurement: the two-term form below is fitted on the measured excess, so its fitted amplitudes absorb the cancellation rather than ignore it.
- **(iii) Graph-relative stationarity**: $\|C_e^{\perp}\|/\|\bar{g}_{\text{src}}\|$ σ-constant, since numerator and denominator share the residual's scale, so that the graph share is flat in σ: a per-route constant, the *floor*.
- **(iv) Data-branch factorization**: $\|B_e^{\perp}\| \approx a_e\,\|\Delta\bar{r}(\sigma)\|$, a σ-independent route amplitude times the *norm* of the cross-grid mean-prediction-residual mismatch of §3.3, asserted to be route-independent and directly measurable as such (§4.4); the measured curve is an *empirical* closure of the mean residual's exact posterior form (Appendix A.3), not an unconstrained fitted shape.

Under (iii) and (iv), with the interference entering as measured and magnitudes read under (i)'s rule, the expansion collapses to the *two-term reduction*,

$$\text{gap}_e(\sigma) \;\approx\; \underbrace{\tfrac{1}{2}\,a_e^2 \cdot \big(\|\Delta\bar{r}(\sigma)\| \,/\, \|\bar{g}_{\text{src}}(\sigma)\|\big)^2}_{\text{data term}} \;+\; \underbrace{\text{RoPE}_e + \text{Resid}_e}_{\text{graph term (the floor)}}, \tag{8}$$

a route amplitude a_e on a measurable mismatch curve plus a per-route constant; the supporting derivations are in Appendix A.4. The reduction is an *effective* law of the cancellation residual, not a bare reading of the shares: its fitted quantities are calibrated on the measured excess, so the amplitude a_e and the floor absorb the near-cancellation of §3.3 rather than model it term by term — which is also the form's stated boundary. Near the amplitude crossing the residual of the cancellation falls *below* the floor the reduction ascribes to the route, a dip no sum of a non-negative data term and a constant can produce; the 768 route's in-window dip (§4.2) is that anticipated signature, and §4.4 reports the cancellation-aware form that prices it at lane parity with the reduction. The floor is written with two parts because a grid change touches two distinct things: the rotary coordinate system, in which every band of the position embedding accumulates phase at a different density across the image (RoPE_e), and the non-positional graph statistics, attention softmax over a different token count and the coarse graph's capacity to approximate the fine graph's computation (Resid_e).

The quadratic power of the data term is fixed by the cosine geometry *locally*; where the measured perturbation turns out large the licensed read is the un-expanded parent form, Eq. 6 evaluated at the same amplitude $a_e\,\|\Delta\bar{r}(\sigma)\|/\|\bar{g}_{\text{src}}(\sigma)\|$ (recovering Eq. 8's data term in the small-mismatch limit), whose data term *saturates* instead of overshooting. The ledger is as follows: Eqs. 6 and 7 are unconditional; Eq. 8 is conditional on (i), (iii), (iv) and on the interference as measured; and the coefficients a_e, $\|\Delta\bar{r}(\sigma)\|$, $\|\bar{g}_{\text{src}}(\sigma)\|$, and each route's floor are measured. One unit convention governs every branch-level magnitude in this paper: writing $h(X) = 1 - \cos(\bar{g}_{\text{src}}, \bar{g}_{\text{src}} + X)$ for the gap that perturbation X alone would realize (so the measured distance is exactly $h(B{+}C)$ by construction), magnitudes are quoted from $h(\cdot)$ or the exact link Eq. 6 only; the quadratic shares $S/\Phi/I$ are read for sign, decomposition, and localization, since at the measured in-window amplitudes the quadratic truncation is out of domain and underpredicts $h(B{+}C)$ (Appendix B).

The estimand forces a debiased instrument. The exact angular link carries a metrological corollary. The estimand is a cosine of population means; any measurement replaces $\bar{g}_a$ with a finite-draw average $g_a = \bar{g}_a + \xi_a$, and the draw noise ξ_a, being uncorrelated with the mean, enters the geometry of Eq. 7 as an orthogonal perturbation share of its own. Every finite-draw cosine is therefore attenuated below its population value, inflating the measured distance by a positive bias, and the bias is arm-dependent: per-draw gradient variance grows as the token count falls, so the inflation is larger on coarser grids and does not cancel in the excess of Eq. 2. The resulting artifact, a positive offset persisting at every σ and growing as the target grid shrinks, matches the signature of the graph floor. The same geometry dictates the remedy's form: uncorrelated with the mean and independent across arms and draw sets, the draw noise cancels in every expected cross term while inflating every squared norm, so to first order the measured cosine factorizes, $\cos(g_{\mathrm{src}}, g_a) \approx \lambda_{\mathrm{src}}\, \lambda_a \cos(\bar{g}_{\mathrm{src}}, \bar{g}_a)$, one attenuation factor $\lambda_a = \left(1 + \mathbb{E}\|\xi_a\|^2/\|\bar{g}_a\|^2\right)^{-1/2}$ per arm. Each factor is measurable in place, since an arm's cosine with an independent redraw of itself, $\cos_{\mathrm{self},a} = \cos(g_a, g'_a)$, has expectation λ_a^2; dividing by the geometric mean of the two self-cosines is Spearman's classical correction for attenuation (Spearman, 1904),

$$\hat{c} \;=\; \frac{\cos(g_{\mathrm{src}},\, g_a)}{\sqrt{\cos_{\mathrm{self,src}} \cdot \cos_{\mathrm{self},a}}}, \qquad \widehat{\mathrm{gap}} \;=\; 1 - \hat{c}, \tag{9}$$

which cancels the finite-draw attenuation of numerator and denominator to first order. The floor claim is therefore readable only through the debiased gap $\widehat{\mathrm{gap}}$; §4.1 instruments Eq. 9 and validates it.

4 Experiments: both accounts, scored

This section scores both accounts of §3 against one measured object: debiased demotion-gap curves over the full σ axis, on routes neither account saw. We build the instrument (§4.1), read the per-route map its curves support (§4.2), and then score both accounts against the same curves: the spectral account at the boundary and the curve level, the gradient-perturbation account term by term (§4.4). Each subsection states the prediction it discharges against criteria frozen before the runs.

4.1 The instrument

Model and data. All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025a), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier-$e$ image occupies roughly $(e/16)^2$ latent-patch tokens (~4,100 at 1024, ~3,000 at 896, ~2,160 at 768, ~1,000 at 512). The probe adapter is a plain LoRA (Hu et al., 2022) checkpoint trained at native tiers: low-rank adapters are the dominant fine-tuning vehicle for models at this scale, and the trainer of §5 trains the same class.

The gradient probe. For each image and each of B σ-bins we accumulate adapter gradients over D stratified noise draws per arm and compare arms by cosine similarity of the flattened accumulated gradients. The verdict run uses $N = 40$ images and $D = 12$ draws per bin, on a segmented grid of 14 bins in $(0, 1)$, dense below $\sigma = 0.1$ and above 0.9, plus a σ=1 endpoint bin, with deterministic kernels. The arms are: a **floor**, two independent draw sets at native resolution, the redraw null; the **reenc** control of §3.1; one **demote** arm per edge e; and, on debiased runs, per-arm **self-floors**, a second independent draw set for every arm. The σ=1 endpoint bin is a distinct probe *mode*: its input is pure ϵ, so the input-mediated data term vanishes by construction. An **x-zero** companion mode removes the image from input *and* target entirely. Both modes read only at the endpoint, since with the $(1{-}\sigma)x$ term absent the lower bins are off-manifold (Appendix B), and together they carry the floor measurement of §4.4. The gap is in cosine units, a scale-free mismatch *fraction*, with bin-mean SEM 0.01–0.07 at N=40. Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. Every bin-mean curve must pass a split-half reliability check over images, a discipline imposed after an earlier failure in which per-image rankings from the same gradients had reliability indistinguishable from zero.

Finite-draw attenuation and the debiased estimator. The arm-dependent attenuation of §3.4 is substantial at our operating point — the natural estimate of Eq. 1 also sets a redraw floor built from *two* finite-draw estimates against demote arms contributing one — and we observed it directly: a first-order variance estimate predicts spurious iso-direction endpoint gaps of $\approx 0.02/0.05/0.15$ for 896/768/512, and the uncorrected 1024→896 endpoint gap indeed decays as $+0.100/+0.035/-0.016$ at $D = 4/8/16$, a clean c/D decay (Appendix B). Two corrections, used together, remove this artifact. First, *self-floors*: every arm runs a second independent draw set g'_a, restoring the floor's two-estimate structure to every arm and supplying the per-arm self-cosines that Eq. 9 consumes; the floor is precisely the source arm's self-cosine, $\cos_{\text{floor}} = \cos_{\text{self,src}}$, and the debiased gap $\widehat{\text{gap}}$ is computed per image and bin. Second, *draw-count extrapolation*: the attenuation is the c/D decay just measured, so fitting $\text{gap}(D) = \text{gap}_\infty + c/D$ over nested draw subsets recovers the draw-limit gap directly. The two are mutually checking, and the check passes: debiased fits come out D-flat ($|c| \le 0.05$ on the 512 route, against $c \approx +0.29$ uncorrected), and the nested sweep extrapolates the native redraw floor to a self-cosine of 1.005 (bootstrap 95% CI $[0.994, 1.016]$). That is, the population per-bin native gradient is a *fixed direction*, and the entire redraw floor is finite-draw noise; remaining validation detail is in Appendix B.

Margins and resolution. A safety verdict needs two numbers, stated separately: a margin, the largest excess considered practically equivalent and a property of the question, and the instrument's resolution, which sets only the power available to decide it. We fix the strict margin at $\varepsilon = 0.02$, comparable to the re-encode control's own confidence half-width (Table 8); a verdict underpowered there is stated at the smallest margin its bounds do clear. A route is *safe at margin ε* in a bin where the one-sided non-inferiority bound clears it,

$$\text{UB}_{95}(\overline{\text{gap}}) \;<\; \varepsilon, \tag{10}$$

and a *window* claim over several bins requires the simultaneous (Bonferroni-corrected) bounds to clear it in every bin. For a route whose true excess is zero the paired estimate $\overline{\text{gap}}$ fluctuates with standard error $\text{SE}(N, D, \sigma\text{-bin})$, so $\varepsilon^* = 1.645\,\text{SE}$ is the *verdict resolution*: a cell with $\varepsilon^* > \varepsilon$ is *underpowered* at the margin, and a crossing there is reported but not counted as a safe verdict. At the verdict grid's operating point of N=40 and D=12 per bin, the bin-level ε^* is 0.02–0.07, so window verdicts below are stated at the margins they do support (0.09 for the windows of §4.2); endpoint-mode draw sweeps, with D up to 64, reach $\varepsilon^* \approx 0.01$–$0.02$, powered for the strict margin. All verdict maps report the *paired per-image excess* $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$, trimmed of non-finite and $|\text{gap}_{e,i}| > 1.5$ values, which cancels the shared per-image draw structure and stays stable at mid-σ where the unpaired estimator is not; finite-draw cosines are compiler-kernel-path sensitive, so all pairings are computed within one run, and the same discipline governs the pooled cross-store vector reads of §4.3, which are licensed only between runs sharing one numerical environment. Every verdict-carrying number in the main text is debiased; the iso-severity, tier-transfer, and positional-interpolation probes of §4.4 remain raw-estimator reads, flagged where cited (§6).

4.2 The measured (route, σ) map

We measured the map directly: the measured (black) debiased gap curves of Fig. 2 carry the verdict, read per bin against the gray $\pm\varepsilon^*$ band per Eq. 10 (all three routes on one axis, with the estimator context, in Appendix Fig. 13; per-bin numbers in Appendix Table 7); §4.4 scores both accounts on the same curves. Throughout, *safe* means the verdict of Eq. 10 on the *per-example* object: the sense in which a demoted gradient is indistinguishable from a native one. Route by route: 1024→896 (floor 0.02–0.04) is safe on the window $\sigma \in (0.5, 1)$ at margin 0.09, simultaneously over the seven window bins; above $\sigma = 0.94$ the bins are underpowered at margins below ≈ 0.05 and the σ=1 endpoint carries a small real gap (below), so the safe condition is $\sigma \in (0.5, 0.94]$ (§6). 1024→768 and 896→768 (floors 0.06–0.09) are safe on no window at the strict margin, but 1024→768 retains a narrow low-excess window $\sigma \in (0.65, 0.95)$, with paired excess +0.02–0.04 per bin, safe at margin 0.09 simultaneously over its four window bins (0.072 pointwise). That excess sits *below* the route's own floor — the anticipated signature of the near-cancellation (§3.3): the window is where the route's branch amplitude ratio comes closest to balance while the anti-alignment stays deep (§4.3); it is the window §5 stacks as a second route. Any route to 512 (floor ≈ 0.30) is *unsafe* in all fifteen bins, the lower bound clearing the margin in every one. Debiasing itself moved verdicts in both directions: roughly half the 768 route's high-σ plateau was estimator bias, softening "unsafe at every noise level" into the narrow window just stated, while the 896 endpoint shows a small *real* gap that single-estimate variance had buried ($+0.034 \pm 0.011$ in the verdict grid's endpoint bin; +0.019 $[+0.010, +0.030]$ in the draw-limit sweep that §4.4 scores).

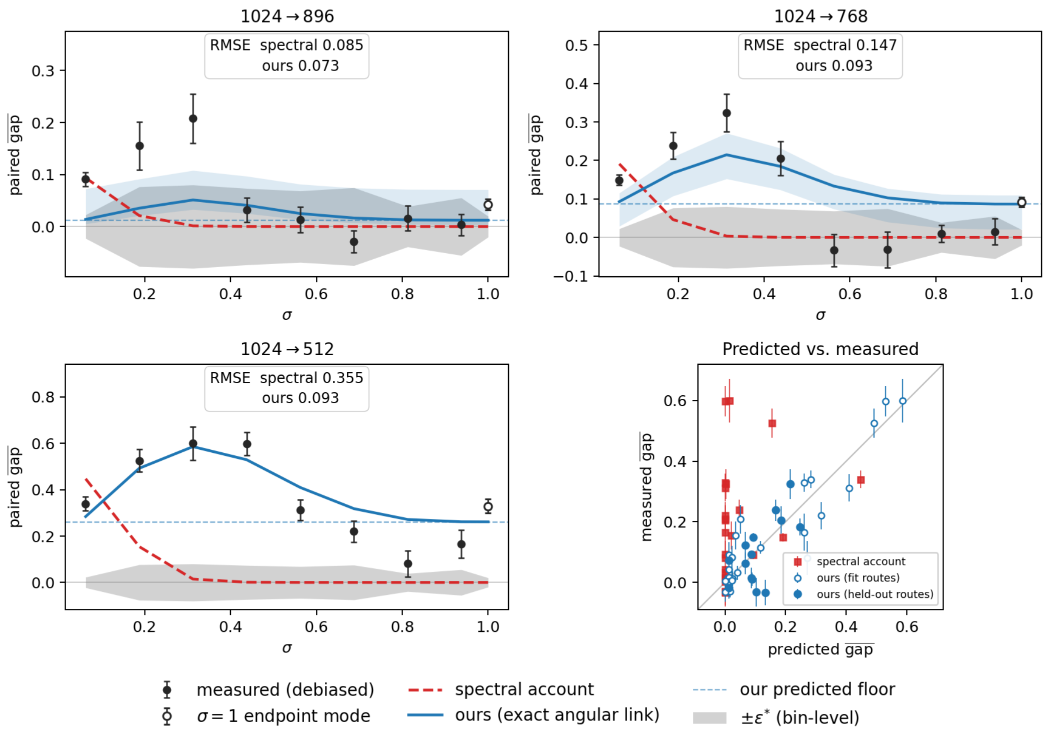

Figure 2: **The measured map, and both accounts on the same axes.** Per route, one set of measured debiased gap curves (Eq. 1, black; open markers detach the σ=1 endpoint bin, a distinct probe mode) carries the verdict of §4.2, read per bin against the gray band, the bin-level $\pm\varepsilon^*$ verdict resolution (§4.1). The same axes carry both predictions of §3: the spectral account (red dashed), transported through the residual→gradient bridge at the measured anchor tolerance, the whole family scoring alike (Appendix Fig. 11); and the gradient-perturbation account under the exact angular link (blue, Eq. 6; three-form comparison in Appendix Fig. 6), with each route's predicted floor (blue dashed; dashed-to-solid is the data term) and a 95% full-pipeline bootstrap band (B=1000; Appendix B).

4.3 The measured geometry of the perturbation

Before either account is scored, we report what the branch decomposition of Eq. 5 measures when both legs are resolved as vectors. The instrument is the probe of §4.1 run with three additional arms, realizing the repromote construction of §3.3 directly: $B = \bar{g}_{\text{rp}} - \bar{g}_{\text{reenc}}$ (the data leg, re-referenced against the control so that the VAE round trip is netted out; the two referencings agree to ~4% wherever the leg carries signal, Appendix B) and $C = \bar{g}_{\text{dem}} - \bar{g}_{\text{rp}}$ per (route, σ-bin), with second moments from cross-draw-set products only and per-leg reliability gates (rel $\geq$ 0.5); magnitudes are quoted in the $h(\cdot)$ units of §3.4 (Appendix B details the arm constructions). Breadth, stated up front for each claim family: the cancellation facts below are measured at every (route, σ) of the operating adapter's verdict grid and replicate, under criteria frozen before the runs, on two further LoRA adapters of the same base model — trained on disjoint data along a designed style axis — at every bin of the 768 window (§6 states this scope in full; 896 replication remains an open cell). The axis-field facts are narrower: one adapter, three vector stores, the standard probe corpus.

The near-cancellation. Fig. 3 shows $h(B)$, $h(C)$, and the realized $h(B+C)$ per σ on each route. At every measured (route, σ) both branches are individually far larger than the realized gap — on the 896 route the data branch alone would realize 104–112% of the measured plateau — and the realized curve sits far below both: the cancellation removes ~70–77% of the summed branch cost on the safe routes $(1 - h(B+C)/(h(B)+h(C))$ averaged over reliable bins: 0.77/0.73/0.63 for 896/768/512), and at the 768 window center the realized gap is ~25–30% of the graph branch's own angular cost. The anti-alignment behind it is deep everywhere, $\rho \in [-0.96, -0.63]$, pooling to $\bar{\rho} = -0.91$ over the gated verdict bins. The crossing geometry of §3.3 reads directly off the amplitude columns: on 896,

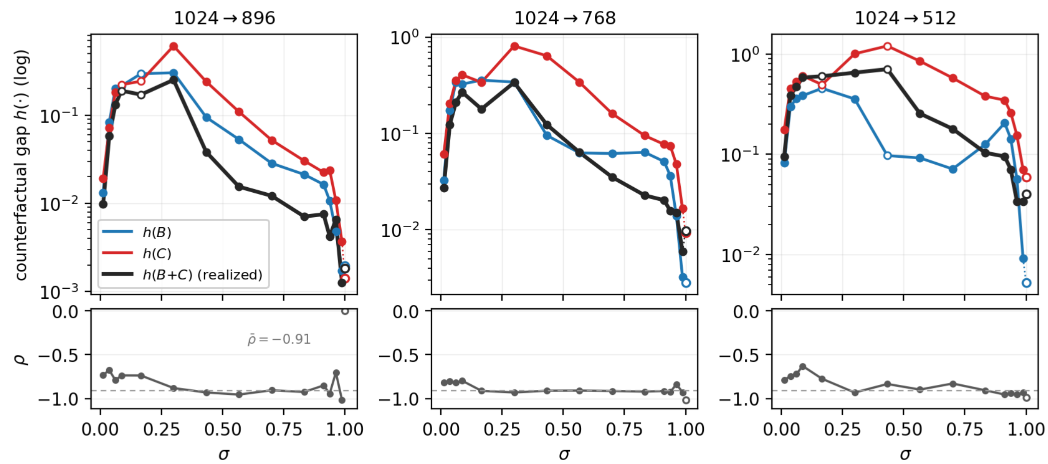

Figure 3: **The gap is the residual of a near-cancellation.** Per route, the exact counterfactual angles $h(B)$ (data branch alone), $h(C)$ (graph branch alone), and the realized $h(B{+}C)$ (= the measured demotion distance by construction), by σ (reenc-referenced vector probe, N=40, D=12; gate-failing bins hollow, σ=1 endpoint detached). The realized curve sits far below both branches at every measured condition; the lower strips give the debiased branch anti-alignment $\rho = \cos(B^{\perp}, C^{\perp})$ per bin, with the pooled $\bar{\rho} = -0.91$ dashed.

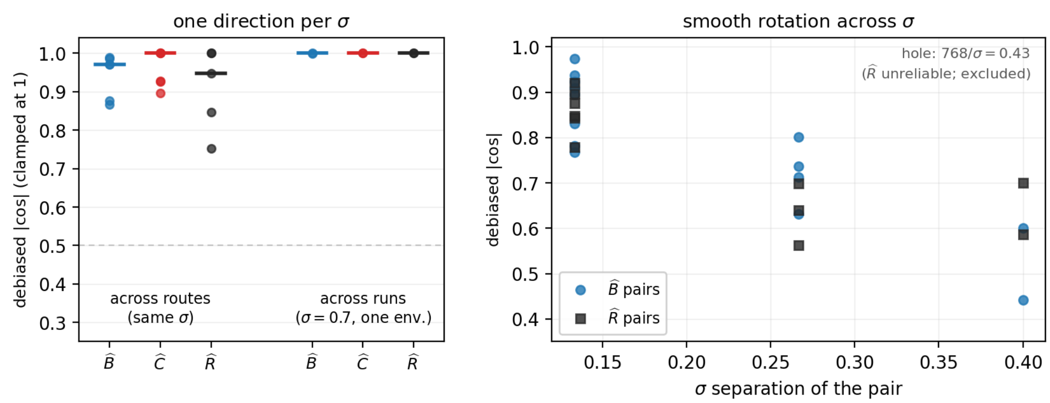

Figure 4: **The residual is a σ-indexed axis field.** Debiased pairwise cosines between pooled directions across conditions, by family: across routes at fixed σ, across two independent runs at $\sigma = 0.7$ (one shared numerical environment; §4.1), and across σ within one route, plotted against σ separation. One direction per σ, shared across routes and runs, rotating smoothly with σ; the recorded reliability hole at $768/\sigma = 0.43$ is marked.

$|B^{\perp}|/|C^{\perp}|$ passes through 1 within one bin of the measured mid-σ cliff ($\sigma \approx 0.43$–0.57); on 768 the fine grid shows no in-window crossing — the ratio rides at 0.76–0.88 through the window, closest to balance exactly where the dip sits — so the 768 window is the crossing geometry approached from below rather than crossed. The unsafe route completes the picture: 512 keeps the same deep angle (mean $|\rho| \approx 0.86$) while its amplitude ratio never reaches balance — what fails on an unsafe route is magnitude matching, not the cancellation itself.

Where the anti-alignment lives. Four targeted reads locate it. *It is born in the pull-back through* $J^{\top}$: the same decomposition applied at the residual level reads only $\rho_r = -0.12$ to -0.22 over the verdict bins, against -0.83 to -0.96 in gradient space; a Gaussian spectral closure derives that weak residual-level seed in sign, shape, and route ordering (missing its level ~1.5–2×), so the depth of the anti-alignment is a property of the compute graph's pull-back, not of the data statistics alone. *It is depth- and type-uniform*: resolved over the network's 28 blocks and 15 module types, every gated cell reads deep (block-level ρ_ℓ median ≈ -0.93; all module types -0.8 to -0.97 where gated, cross-attention and MLP included), with interaction magnitude proportional to branch energy at every depth — no excess interaction anywhere, including the query/key rows rotary touches directly. *It is locally enforced*: read per (depth × module-type) cell, the gated-cell median ρ_{cell} is -0.90 to -0.96 with interquartile range at most 0.11 at every verdict σ on both routes — the pooled constant is not carried by a few cross-cell modes but holds pointwise in parameter space. *And at noise-dominated bins it is partially phase-borne*: re-running the demoted graph with its rotary phases interpolated to match native geometry rotates the graph branch ($\cos(C^\perp, C^\perp_\pi) = 0.50$–$0.67$ on 768), halves the anti-alignment ($\rho \to -0.34$ to -0.60), and *doubles* $h(C)$ — with phases aligned the graph branch stops magnitude-matching the data branch, so the phase geometry is part of what lets the branches cancel. The amplitude of this phase response concentrates in the adaptive-norm band (86–87%), while the direction rotation is carried globally. Finally, the whole structure is *operating-point invariant*: re-running the residual-level probe bit-matched with the adapter merged moves neither branch (per-image raw cos ≥ 0.93 at every gated bin), extending the checkpoint-controlled replication of Appendix F to the branch level. Together these reads license quoting $\bar{\rho}$ as one constant.

The residual is a σ-indexed axis field. The near-cancellation leaves a residual direction $\widehat{R} = \widehat{(B+C)}$ per (route, σ), and its legs $\widehat{B}$, $\widehat{C}$ likewise; comparing them across conditions (debiased cosines, Fig. 4) shows the residual is not condition-noise. At fixed σ the direction is shared: across routes, median $|\cos| = 0.97$ for the legs and 0.95 for $\widehat{R}$ — 896 and 768 demotion push the adapter along the same line — and across two independent runs the debiased cosine is ≈ 1. The across-run read carries a scope qualifier: finite-draw directions are compiler-kernel-path sensitive (§4.1), and the two stores compared share one numerical environment; cross-environment vector comparability is not claimed. Across σ the axis rotates smoothly — adjacent bins 0.89–0.97, decaying monotonically with σ separation to 0.44–0.60 over the widest span, with no sign flips; a spherical-interpolation check predicts held-out mid-span directions at median $|\cos| = 0.96$, confirming smoothness as a description (a parametric rotation law was tested and is *not* claimed: the best candidate performed worse than interpolation). Two honest boundaries. First, the rotation is matched-angle, not planar: the axis rotates through approximately the same angle as the native gradient direction $\hat{g}(\sigma)$ itself, but in its own plane — transporting a bin's direction by the anchor's measured rotation buys nothing (median gain -0.0005) — so the natural reading that the residual simply co-rotates with the σ-conditioned native gradient is retired. Second, reliability: the pooled residual direction passes its reproducibility gate at 11 of 12 verdict conditions; the exception is $768/\sigma = 0.43$, and mid-window 768 is generally where the direction is least reproducible — the axis-field claims carry this hole explicitly (supporting panels in Appendix D). A decomposition of the residual gap into its two knobs says which one matters: setting $\rho \to -1$ at measured amplitudes would remove 70–100% of the residual at every gated condition, while matching amplitudes at the measured angle removes at most 45% — the residual gap lives in the incomplete angle, not in amplitude mismatch.

Closures. Four adjacent doors are closed by measurement, and we report them as results. (1) *The account's data term is not derivable from the geometry at the estimand level*: replacing the measured mismatch curve $x(\sigma)$ with the ledger's own (measured *or* closure-predicted) branch-amplitude curve fails the held-out lanes at 4–6× the measured lane's RMSE — and measured and derived amplitudes fail *equally*, so the failure is an estimand mismatch between the branch legs and the account's $x(\sigma)$, not a modeling miss. The same read closes re-deriving any ledger term as an objective correction. (2) *Damping-form levers are dead at probe level*: an exact one-step counterfactual sweep of adaptive-norm-band damping on the committed stores finds the optimal damping $\approx$ identity almost everywhere; the entire sweep buys ≤ 0.006 of gap closure at 13% signal cost. (3) *A fixed subspace is refuted at lever level*: projecting out a single shared residual subspace is actively harmful (signal retention 0.75–0.82); any projection-style object must be per-σ-bin. (4) Per-sample exploitation of the geometry remains gated on its own pre-registered chain and is not claimed: every statement above is about pooled directions at this operating point.

4.4 Both accounts, scored

Both accounts are scored against one measured object: the debiased gap curves of Fig. 2, read as §4.2 has just set out. The spectral account is scored first, at the boundary and then at the curve level; the gradient-perturbation account follows, term by term.

The spectral account. The spectral account's prediction is monotone: the predicted gap is concentrated at low σ and vanishes at high σ, once noise drowns the destroyed band (equal-power crossovers $\sigma \approx 0.14/0.15/0.20$ for 896/768/512, zero predicted gap above). The measurement disagrees in both regimes: the transported curves are near zero precisely where the measured curves peak (RMSE 0.147 at 768, 0.355 at 512), and above the crossover the floor persists (committed-region RMSE 0.027/0.049/0.218). Neither read depends on the tolerance or the residual-shape choice in the transport (within 0.01 RMSE; Table 5, Fig. 11, Appendix E).

The data branch: a route-uniform residual mismatch. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ of §3.4 per grid and σ shows a strong, monotone σ-shape (cross-grid excess $0.89 \to 0.36$ in relative L_2 from σ=0.125 to 1) that is the *same for every route* within ±0.02, across routes whose gradient floors span 0 to 0.3 (Appendix Table 12). This result refutes the residual-level prediction of §3.2: with no cross-frequency coupling, that mismatch is confined to the destroyed band and ordered by its energy, which differs substantially across these routes. Route-uniformity implies that the real mismatch is carried by cross-band structure the diagonal model cannot express, and that all route identity lives in J. The same contrast reads on assumption (iv): the measured route-uniformity is the shape factorization that assumption asserts, established here directly rather than assumed. The decay is smooth, Wiener-like shrinkage with no hard gate at the spectral crossover. How this monotone mismatch, read over the U-shaped total gradient norm, yields the measured mid-σ-peaked curves is a post-hoc consistency read deferred to Appendix B.

The graph branch: the floor exists. At the $\sigma = 1$ *endpoint* the input-mediated data term is zero by construction (§3.3); the debiased draw-limit endpoint floors are +0.019/+0.056/+0.304 for 896/768/512 (Appendix Table 8), and the 512 value clears the pre-registered confirmation bar ($\widehat{\mathrm{gap}}_\infty \geq 0.15$) on 12 of 12 probe images individually. Three companion probes harden this number without adding to it. The *x-zero* probe reproduces the endpoint floors at every route (Table 8); a *target-strength sweep* (target $\epsilon - \alpha x$, $\alpha \in [0, 1]$) finds the paired debiased excess α-flat, discharging the survivor question of §3.3 — the target-mediated gradient is real but demotion's change to it lands along $\hat{g}_{\mathrm{src}}$, to which the angular estimand is blind (Eq. 6); and a forward-only probe of the model's caption-conditioned *prior* dissociates the prior from the floors' route ordering, placing the ordering in J. The floor's split into $\mathrm{RoPE}_e + \mathrm{Resid}_e$ (Eq. 8) is then read by one origin-side intervention: evaluating the demoted grid at positionally-interpolated fractional rotary coordinates, which matches its relative phase geometry to native, *erases* the 768 endpoint floor and removes ~30% of the 512 floor, the in-band 896 control untouched (a raw-estimator probe; §6). Re-anchored against the debiased floors, $\mathrm{Resid}_{768} \approx 0$: what survives the coordinate fix is the harsh route's $\mathrm{Resid}_{512} \approx 0.2$ bulk, a genuine non-positional graph residue, to which the capacity governor below attaches. (The erasure itself is no training lever; the frequency-selective banded alignment of §5 is the form that survives in-window.) Constructions, anchors, the parallel-landing decomposition behind the α-flatness, and the floor ledger's full numbers with their per-block depth localization are in Appendix B.

The governors: ratio sets the amplitude, absolute size sets the floor. With the floor isolated, what orders the routes? Two probes give a clean division of labor, and together they decide the equal-ratio disagreement, limit (b) of §3.2. *Absolute size, not ratio, sets the floor*: re-run one tier down, the route 896→768 fails its pre-registered bar with a high-σ residual ~2× the 1024→896 plateau despite a near-identical downscale ratio, while the two routes sharing the ~2,160-token target grid, 1024→768 and 896→768, land on the same floor despite different ratios (raw-estimator read, the separation confirmed by the debiased same-grid floors; Appendix Table 11). Near-equal ratio with a different target gives a different verdict; the same target with a different ratio gives the same floor. This refutes any pure ratio rule for the floor, and with it the spectral account's only degree of freedom. Moreover, the floor grows monotonically as the *target* grid shrinks (0.02–0.04 / 0.06–0.09 / ≈ 0.30 at target ~3,000/2,160/1,000 tokens), consistent with the reading that the floor measures how well the coarse graph approximates the fine graph's computation. *Ratio, not absolute size, sets the amplitude*: a synthetic iso-severity route 1280→1120, with edge ratio exactly matched to 1024→896 at 1.6× its absolute target size, reproduces the 1024→896 curve bin-for-bin, crossover included (raw-estimator read; Appendix Table 10). Note what survives of the spectral account here: ratio *is* the right governor for the data term; its error is not its governor but its scope, mistaking one term for the whole gap.

Table 1: Weight-space footprint: $\cos(\Delta W_{\text{arm}}, \Delta W_{\text{native}})$ at matched training seed, mean ± SD over three seed pairs, on two single-artist corpora (60/15 training images). Training uses deterministic kernels, including FlashAttention's deterministic backward (Dao et al., 2022) (an identical retrain reads 1.000); the non-det. row retrains native at the same seed *without* deterministic kernels (one twin pair per corpus), so it reads run-to-run kernel nondeterminism alone. *late*: demotion only in the last 75% of steps; the stacked row adds the 768 low-excess window as a second route (1024→768 on $0.65<\sigma<0.95$, last 50%). **Bold**: closest endpoints.

arm (vs. native, matched seed)	corpus A	corpus B
native, same-seed non-det. retrain	0.264	0.427
896, $\sigma>0.5$, aligned	0.365 ± .020	0.432 ± .026
+ late	**0.753** ± **.019**	**0.770** ± **.009**
+ late, stacked 768	**0.753** ± **.017**	**0.771** ± **.009**
896, every σ	0.183 ± .009	0.236 ± .009

Table 2: The proposed step, in full. Lines 3–5 (colored) are the entire change: the gate σ^*=0.5 and the route are read off the measured map (§4.2), line 5 is the optional refinement, and lines 1–2 are only a reordering of a standard step, such that σ is drawn before the latent is fetched.

one LoRA training step

1 $\sigma \sim p_{\text{train}}$
2 $x \leftarrow$ `native latent of` y
3 `if` $\sigma > \sigma^*$ `and` y `on-route:`
4 $\quad x \leftarrow$ `demoted latent of` y
5 $\quad$ `rotary` $\leftarrow$ `banded(`σ`)`
6 $\epsilon \sim \mathcal{N}(0, I)$ `at shape(`x`)`
7 $z \leftarrow (1-\sigma)x + \sigma\epsilon$
8 $v \leftarrow \epsilon - x$
9 `step on` $\|\hat{v}_\theta(z, \sigma, c) - v\|^2$

The cancellation-aware form, at lane parity. Since the geometry of §4.3 suggests an operational form of its own — the exact link Eq. 6 evaluated on *both* shares with the interference at the pooled constant $\bar{\rho}$ — we scored it under the identical held-out protocol as Eq. 8 (same mismatch curve bit-identically, same governors, same lanes and bootstrap; only the link differs). It wins exactly one safe lane: held-out 768, RMSE 0.078 against the additive form's 0.099, and from held-out parameters alone it reproduces the measured non-monotone shape and places the in-window dip at $\sigma = 0.67$ — the one structure the additive form cannot express. It loses the other safe lane (leave-one-out 896: 0.086 vs. 0.077) because its graph-side constant collapses on the 1280 tier and the cross-tier constant law does not transfer, and it breaks the guard tier the additive lane already covers (1024 held-out: 0.199 vs. 0.046). We therefore report it at lane parity and do *not* adopt it: the additive lane keeps Fig. 2, and the reading is that the interference geometry transfers through governors on the tier that licensed $\bar{\rho}$ while its amplitude law across tiers remains open (demonstration panel and full lane numbers in Appendix D).

5 σ-CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

As an applied example, we introduce the measured-safe corpus route (1024→896) into a LoRA (Hu et al., 2022) trainer as an opt-in path, read off the map as measured, with one stated liberty: the gate is one-sided ($\sigma > \sigma^*$), so demoted steps also land in the underpowered upper tail $\sigma \in (0.94, 1]$ (§6). The intervention is deliberately small: three lines inside an otherwise standard flow-matching step, and no change to the objective, the optimizer, or the noise density (pseudocode in Table 2). We exercise it on a fixed-step grid: gated, late-scheduled (alone and with the 768 low-excess window stacked as a second route), and gate-removed arms × 3 training seeds × two single-artist illustration corpora of contrasting style, one flat and graphic, one densely rendered (corpus A, 60 training images; corpus B, 15), 480 optimizer steps each, run with deterministic kernels (including FlashAttention's deterministic backward, Dao et al., 2022) and paired RNG so every arm sees identical draws. At fixed steps the always-gated arm realizes a −14.6% training-time saving over native training on our corpora; the late-scheduled arms demote only part of the run and save proportionally less.

σ-first draw. The trainer draws each batch's σ *before* fetching latents, from the unchanged training density. When every sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is replaced with a cached demoted-grid counterpart produced by the probe's own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream derives from the substituted latent. Off-route images always train native, as does validation.

Banded rotary alignment, σ-gated. Motivated by §4.4, a frequency-selective alignment is applied on demoted forwards only. It uses YaRN-style bands: a full positional-interpolation stretch for rotary

Figure 5: Visual counterpart to Table 1: matched (prompt, noise-seed) renders from the arms' endpoints on a corpus-B display prompt, 20 steps; columns are four of the table's arms (the unscheduled gated arm is omitted), rows the three training seeds. Moving along a row (arms, seed fixed) changes the render on the same visual order as moving down a column (training seed, arm fixed); the gate-removed arm (rightmost) drifts the most, matching its lowest cosine in Table 1. The positive prompt is given in Appendix G.

bands with few rotations across the demoted extent, native spacing for high-frequency bands, a ramp between, and thresholds scheduled by a sigmoid gate in σ following Zhao et al. (2026). Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs, erasing the static variant's low-σ liability ($+0.064 \rightarrow +0.033 \pm 0.025$) while preserving its high-σ gains (paired -0.05 vs plain demotion at $\sigma \geq 0.59$).

Weight-space endpoint footprint. Table 1 compares the trained adapters directly in weight space, as the cosine between each arm's endpoint ΔW and its matched-seed native twin's. Determinism makes the read exact: an identical retrain reproduces its twin at 1.000, and the same retrain with non-deterministic kernels reads 0.264/0.427 (corpus A/B), run-to-run kernel nondeterminism alone. Against that reference, demoting at *every* σ falls below the non-deterministic retrain (0.183/0.236), while demoting only above the gate (σ>0.5) matches or exceeds it (0.365/0.432). Late scheduling, demotion only in the last 75% of steps, brings the endpoint very close to the native weights (0.753/0.770), and, as the map's low-excess read of the 768 window predicts, stacking the 1024→768 route onto the late rule leaves the cosine essentially unchanged (0.753/0.771). The visual counterpart (Fig. 5) confirms the read: matched (prompt, seed) renders from the late-scheduled arms are nearly identical to native.

6 Limitations

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint is the designated bound on how far the map generalizes, as is re-probing 1280→1024 after fine-tuning at the 1280 tier. The probe pool's relationship to the adapter's own fine-tune set was measured rather than conceded: a controlled 2×2 factorial (two LoRAs trained on opposite style clusters under frozen membership manifests, probed on both) replicates the map's shape on both checkpoints with the adapter×probe-style interaction below instrument resolution, while the absolute redraw-floor level is checkpoint-dependent (Appendix F). The cancellation geometry of §4.3 was additionally replicated on the same two controlled checkpoints under criteria frozen before the runs: at every bin of the 768 window both adapters pass the frozen cancellation criteria (negative interference, $\rho \leq -0.5$ — measured −0.89 to −0.98 — and a realized gap below either branch alone), so the cancellation-rung breadth is three adapters of this base model, trained on disjoint data along a designed style axis, at 768 across the window; 896 replication remains the stated open cell. Whether the cancellation axis is a property of the base model or of the adapter stays open: gradients with respect to different adapters' parameters live in mostly non-overlapping frames, so a raw cross-adapter direction read cannot separate the two (it shows no evidence of a privileged shared axis above frame overlap, and none against structure below it). The axis-field claims of §4.3 make no cross-adapter statement at all: their scope is one adapter, three vector stores, one probe corpus.

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed by the exercise grid's weight-space read at 480 steps rather than production scale.

Account resolution and domain. The account predicts held-out routes at ~0.07–0.09 RMSE, not within ε^* (the held-out protocol, its pre-registered gates, and the structural failures it exposes are quantified in Appendix C); its $A(\text{ratio})$ governor rests on two distinct ratio values; the reduction's domain excludes the 768 mid-σ window, the crossing neighborhood §3.4 states as the effective form's boundary up front (§4.3 measures it; the cancellation-aware form prices it but is not adopted, §4.4); and the floor law's functional form is unidentified at our operating points. The graph-floor reading of the endpoint is also estimand-specific: the target-mediated perturbation is real at the vector level and lands parallel to the native gradient (Appendix B), so a magnitude-sensitive estimand, such as an un-normalized optimizer, could apportion the endpoint gap differently than the angular read does.

Instrument resolution and coverage. "Safe" is one-sided non-inferiority against a fixed margin ($\varepsilon = 0.02$ strict; the map's windows are currently safe at 0.09 simultaneous, ≈ 0.07 pointwise), with the verdict resolution ($\varepsilon^* = 0.02$–0.07 per bin; ≈ 0.01–0.02 at the endpoint sweeps) setting the power available, not the margin. True gaps below ε^* are undetectable by construction, σ^* boundaries are localized only to a bin, and every failure to reach a safe verdict on a window is (N, D)-relative: a larger campaign could establish (or refute) the 768 high-σ window at the strict margin. The upper tail $\sigma \in (0.94, 1)$ of the safe route is underpowered: its two non-endpoint bins resolve only margins above ≈ 0.05, and the σ=1 endpoint carries a small real gap ($+0.034 \pm 0.011$); an upper-tail sweep at higher draw count is the designated probe to close the region, and until it lands the safe condition is $\sigma \in (0.5, 0.94]$. Verdict stability under the per-image trim rule of §4.1 has not been separately audited. Vector-resolved directions carry one further instrument bound: finite-draw directions are compiler-kernel-path sensitive, so every cross-arm and cross-store pairing in §4.3 is computed within one run or between runs sharing one numerical environment (§4.1); cross-environment vector comparability is not claimed. The debiasing campaign covered the corpus verdict grid; the 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention probes remain pre-debiasing reads, and the 1280-tier curves enter the held-out validation on that basis.

Outlook. The geometry of §4.3 names, but does not test, a population-level lever family: the closures there dictate that any exploitation must be σ-local (a fixed subspace is refuted at lever level) and angle-targeted (the residual gap lives in the incomplete angle) — for instance a per-σ-bin residual-direction guard restricted to the conditions where the pooled direction is reliable, or an explicit resolution conditioning of the pathway that carries the phase-response amplitude. These are licensed by the measurements and unrun; no lever result is claimed or implied, and nothing in this paper depends on them.

7 Conclusion

When does training on downscaled images yield the same gradient direction? We answered by resolving what demotion actually perturbs — two individually large, strongly anti-aligned branches whose near-cancellation leaves the measured gap as its residual — by stating the effective two-term account of that residual, a ratio-governed data term plus a σ-independent, token-count-governed graph floor, and by building the debiased estimator the question requires, since naive finite-draw estimation manufactures the very floor signature under test. Measured under that instrument, the spectral answer of the inference-side literature (safe once noise masks the frequencies the coarse grid loses) models only one part of one branch of what demotion perturbs: the network function itself is grid-calibrated, partly through its rotary coordinate system and partly through an irreducible dependence of the computation on token count. The gradient-perturbation account is predictive, not merely descriptive, reaching held-out routes at ~0.07–0.09 RMSE against the spectral family's 0.147–0.355, and its map licenses a selective trainer: on our model, 1024→896 above a one-sided gate ($\sigma > 0.5$), realizing -14.6% training time at fixed steps with a weight-space endpoint within kernel-noise reach of a native retrain, rising to cosine 0.75 when demotion is late-scheduled at proportionally reduced saving, with the low-excess 1024→768 window stackable onto the late rule at no endpoint cost. We offer the account and its instrument as the inexpensive, general test that any future "train part of the time at lower resolution" proposal should pass before it is believed.

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A Notation and derivations

A.1 Notation and estimand fine print

Arm labels versus route subscripts. One notational convention holds throughout. Arm labels (src, reenc, rp, dem) mark *whose* gradient, latent, or Jacobian an object is; the subscript on every route-level scalar is the *route*, written in full ($d_{e_0 \to e}$) where several sources appear, and abbreviated to the target edge alone wherever the source edge is fixed by context, as it is in almost every measured statement; the abbreviation is an edge value, never an arm label, so it always instantiates numerically (d_e, Φ_{1120}, Resid_{768}). The control's distance d_{reenc} (Eq. 1 with $\bar{g}_{\mathrm{reenc}}$ in place of $\bar{g}_{\mathrm{dem}}$) is the one exception that carries an arm label: the control changes no grid and has no target edge. Each table states which object it reports, d_e or the excess gap_e.

The two aggregations. In a batched training scenario Eq. 1 admits two aggregations over a probe set: per-example (cosine per image, then mean; the object every map in the body reports) or batch-aggregate (gradients summed over the batch before the cosine; the object SGD actually follows). These are different estimands, since the aggregation operator is part of the estimand, and no coefficient fitted on one is guaranteed to transfer to the other. At second order the two are related by a batch-size decomposition: writing each image's demotion-induced gradient disagreement as a coherent mean b plus a zero-mean idiosyncratic deviation with covariance Σ_η, the batch-aggregate distance at batch size B is

$$\mathbb{E}[d_B] \;\approx\; \frac{\|P^\perp b\|^2}{2\|\mu\|^2} \;+\; \frac{\mathrm{tr}\big(P^\perp \Sigma_\eta P^\perp\big)}{2B\|\mu\|^2},$$

with μ the mean source gradient and $P^\perp$ the projector off $\hat{\mu}$: an intercept (the coherent share, which never averages out) plus a $1/B$ term. Monotone improvement with B is not automatic, since it would require an iid zero-mean disagreement model we have not verified, but where we have measured the batch object it is the more forgiving of the two: pooled-arm probes at pool size 4 collapse the 1024→896 excess to ≈ 0 at every bin $\sigma \geq 0.625$ and the 1024→768 excess at $\sigma \geq 0.875$, the $1/B$ term dominating there. What the collapse does not bound is the intercept, the coherent drift no batch size removes. A debiased verdict grid with pooled self-floors (pool 4, N=40; paired arm−reenc gaps at $B \in \{1, 4, 40\}$) bounds it: the $a + b/B$ fit's intercept is ≤ 0.02 (1024→896) and ≤ 0.03 (1024→768) at $\sigma \geq 0.44$, but persists over the lower bins (+0.04–0.09, +0.05–0.28, and +0.15–0.60 for 896/768/512) — aggregation confers safety only where the per-example map of §4.2 is already near-safe. The small real 896 endpoint gap (+0.042 in this run's B=1 read) averages out in the aggregate (+0.001).

A.2 Derivation of the branch decomposition (Eq. 5)

Fix a query (image, caption), a σ, and an arm a. Per draw ϵ, the adapter gradient of the $\frac{1}{2}$-scaled loss is the product $g_a(\epsilon) = J_a(\epsilon)^\top r_a(\epsilon)$ of §3.3. The three arms of Eq. 5 — src (native content, fine graph), rp (demoted content, fine graph), dem (demoted content, coarse graph) — are values of one map on the data–graph lattice, and their adapter gradients all live in the shared adapter-parameter space, so the differences are literal and the split

$$\delta g \;=\; \bar{g}_{\mathrm{dem}} - \bar{g}_{\mathrm{src}} \;=\; \underbrace{\big(\bar{g}_{\mathrm{rp}} - \bar{g}_{\mathrm{src}}\big)}_{\text{data branch } B_e} \;+\; \underbrace{\big(\bar{g}_{\mathrm{dem}} - \bar{g}_{\mathrm{rp}}\big)}_{\text{graph branch } C_e} \tag{11}$$

telescopes exactly: Eq. 5 is an identity at every amplitude, with no remainder and no linearization. The lattice's fourth corner, native content on the coarse graph, does not exist — representing native content on a coarser grid *is* demotion — so the path through rp is the only realizable one and the split is unique (Fig. 7).

What each leg collects follows from the mean-factor expansion. Split each factor into its noise mean and fluctuation, $J_a = \bar{J}_a + \tilde{J}_a$ and $r_a = \bar{r}_a + \tilde{r}_a$ with $\mathbb{E}_\epsilon[\tilde{J}_a] = 0$ and $\mathbb{E}_\epsilon[\tilde{r}_a] = 0$; the population mean gradient of arm a is then

$$\bar{g}_a \;=\; \mathbb{E}_\epsilon\big[J_a^\top r_a\big] \;=\; \bar{J}_a^\top \bar{r}_a \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_a^\top \tilde{r}_a\big]. \tag{12}$$

Applying Eq. 12 to the two arms of each leg, with the demoted data's mean factors written as source plus a perturbation ($\bar{r}_{\mathrm{dem}} = \bar{r} + \Delta\bar{r}$, $\bar{J}_{\mathrm{dem}} = \bar{J} + \Delta J$; bare bars denote the source arm): the data branch collects everything the data intervention does at fixed graph — through the residual (its leading term is the pulled-back residual perturbation $\bar{J}^\top \Delta\bar{r}$), through the input-content dependence of the Jacobian at fixed grid, and through the shift of the noise-covariance coupling $\mathbb{E}_\epsilon[\tilde{J}_a^\top \tilde{r}_a]$. The graph branch collects everything the grid change does at fixed data: its leading term is the changed pull-back on the residual, $\Delta J^\top \bar{r}$, and because the graph difference is taken at the *perturbed* data, the second-order coupling $\Delta J^\top \Delta\bar{r}$ rides inside it, along with the grid-driven covariance shift. Relative to a direct two-factor product rule between src and dem, the terms that expansion would have collected as a remainder — the second-order coupling and the covariance differences — appear *inside* the legs, assigned to whichever intervention realizes them along the path; nothing is discarded.

One remark delimits what is and is not computable. When two arms share a grid (as for the re-encoding control, or the src–rp pair) every operation above is literal. Across grids, J_{src} and J_{dem} act on different token counts, so the mean-factor differences ΔJ and $\Delta\bar{r}$ are matrix formulas only under a fixed identification of the two output spaces (e.g. spectral zero-padding of the coarse grid onto the fine grid's bands), and any such identification moves content between the nominal factors while leaving the legs untouched. This is why the paper treats the mean-factor forms as the legs' leading-order *interpretation* and measures only the legs themselves, via the designated probes of §4.4; no cross-grid matrix product is ever computed.

A.3 The mean residual as a posterior operator

This appendix derives the posterior form of the fixed-image mean residual, the exact split of the measured object into intrinsic posterior bias plus model approximation error, and the Gaussian closure referenced in §3.4. Fix a caption c and grid e, write $a = 1 - \sigma$, and let $x \sim p_e(\cdot \mid c)$ be the grid-e clean latent, $z_\sigma = ax + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$.

Bayes field and fixed-image residual. For the squared objective the population-optimal field is the conditional mean of the target $v = \epsilon - x = (z_\sigma - x)/\sigma$:

$$v_e^*(z,\sigma,c) \;=\; \mathbb{E}[v \mid z_\sigma = z,\, c] \;=\; \frac{z - m_e(z,c)}{\sigma}, \qquad m_e(z,c) := \mathbb{E}[x \mid z_\sigma = z,\, c]. \tag{13}$$

Averaging over noise at fixed image ($\mathbb{E}_\epsilon[\epsilon] = 0$, $\mathbb{E}_\epsilon[z_\sigma] = ax$) gives the posterior form used in §3.4:

$$\bar{r}_e^*(\sigma;x) \;=\; \mathbb{E}_\epsilon\big[v_e^*(ax+\sigma\epsilon,\sigma,c) - (\epsilon - x)\big] \;=\; \frac{x - \mathbb{E}_{z_\sigma|x}\big[m_e(z_\sigma,c)\big]}{\sigma}. \tag{14}$$

With $p_{\sigma,e}(z \mid c)$ the noised marginal, Tweedie's identity for the linear interpolant reads $m_e(z,c) = \big(z + \sigma^2\nabla_z \log p_{\sigma,e}(z \mid c)\big)/a$ for $\sigma < 1$, so equivalently

$$\bar{r}_e^*(\sigma;x) \;=\; -\frac{\sigma}{1-\sigma}\,\mathbb{E}_{z_\sigma|x}\big[\nabla_z \log p_{\sigma,e}(z_\sigma \mid c)\big], \tag{15}$$

which makes the missing ingredient explicit: the numerical curve requires the grid- and caption-conditional smoothed score, i.e. a model of $p_e(x \mid c)$. At $\sigma = 1$, where $z = \epsilon$ is independent of x, $m_e(z,c) \to \mathbb{E}[x \mid c]$ and $\bar{r}_e^*(1;x) = x - \mathbb{E}[x \mid c]$. With A_e the alignment operator that places the source-grid residual on the demoted grid (§4.1), the Bayes-level cross-grid mismatch is $\Delta\bar{r}_e^*(\sigma;x) = \bar{r}_{\text{dem}}^*(\sigma;x_{\text{dem}}) - A_e\bar{r}_{\text{src}}^*(\sigma;x_{\text{src}})$.

What the instrument measures. MSE optimality gives $\mathbb{E}[v^* - v \mid z,c] = 0$ and hence a zero residual after a *joint* average over (x,ϵ); it does not zero the fixed-image average $\mathbb{E}_\epsilon[v^* - v \mid x,c]$, which is Eq. 14 and generally nonzero. Writing the trained network as $\hat{v}_{\theta,e} = v_e^* + q_{\theta,e}$, the probe's per-image object splits exactly as

$$\bar{r}_{\theta,e}(\sigma;x) \;=\; \bar{r}_e^*(\sigma;x) \;+\; \mathbb{E}_{z_\sigma|x}\big[q_{\theta,e}(z_\sigma,\sigma,c)\big]\;: \tag{16}$$

intrinsic posterior reconstruction bias, derivable in principle from $p_e(x \mid c)$, plus genuinely model-specific approximation error. The measured $\|\Delta\bar{r}(\sigma)\|$ therefore has definite theoretical status, as an empirical closure of Eq. 14, but is not a pure model-error difference, and no claim in the paper requires it to be one: the two-term reduction consumes the measured curve as-is (assumption (iv)), and the trained-model residual is in any case the exact r-factor that the LoRA gradient $g = J^\top r$ is built from.

Gaussian closure. The simplest nontrivial closure is a full Gaussian model $x \mid c \sim \mathcal{N}(\mu_e, C_e)$ per grid. With $Q_e(\sigma) = a^2 C_e + \sigma^2 I$, the posterior mean is linear, $m_e(z, c) = \mu_e + a C_e Q_e^{-1}(z - a\mu_e)$, and substituting into Eq. 14 collapses (via $I - a^2 C_e Q_e^{-1} = \sigma^2 Q_e^{-1}$) to

$$\bar{r}_e^*(\sigma; x) \;=\; \sigma\, Q_e(\sigma)^{-1}(x - \mu_e). \tag{17}$$

For paired source and demoted latents with covariance blocks C_{ss}, C_{dd}, C_{ds} and $L_s = \sigma Q_s^{-1}$, $L_d = \sigma Q_d^{-1}$, the image-population mean squared mismatch is

$$\mathbb{E}\big\|\Delta \bar{r}_e^*\big\|^2 = \operatorname{tr}\big(L_d C_{dd} L_d^\top\big) + \operatorname{tr}\big(A_e L_s C_{ss} L_s^\top A_e^\top\big) - 2\operatorname{tr}\big(L_d C_{ds} L_s^\top A_e^\top\big), \tag{18}$$

a σ-curve computable from paired *clean latents only*, with the source–demote cross-covariance C_{ds} load-bearing. A *diagonal Fourier C_e* makes Eq. 17 the per-band Wiener shrinkage of §3.2, confining the mismatch to the destroyed band and ordering it by destroyed-band energy: the residual-level prediction that route-uniformity falsifies (§4.4). We tested whether any structured second-order closure suffices, entirely on stored paired clean latents (no denoiser forward): three nested covariance models (diagonal spectral; + within-band channel blocks; + octave-pair cross-band coupling), fitted on held-out-from-probe images and scored on the probe set against the measured curves. All three reproduce the curve's monotone σ-shape (Pearson 0.94–0.97), its route-uniformity, the near-zero re-encoding control, and the σ=1 endpoint, but over-predict the low-σ amplitude by ~40%, failing a pre-registered RMSE bar. Second-order data-only structure therefore recovers the shape but not the level, and the essential missing ingredients are the ones the identity names: caption-conditioning (the closure is caption-marginal), non-Gaussian posterior structure, and the q_θ term of Eq. 16. The paper therefore retains the measured $\|\Delta\bar{r}(\sigma)\|$ as its minimal honest closure. A theory–instrument comparison must also reproduce the instrument's estimand, the split-half-corrected *relative* L_2 distance of §4.1, i.e. $\sqrt{\mathbb{E}\|\bar{r}_d - A_e \bar{r}_s\|^2}$ normalized by the mean of the two arms' root-mean-square norms, rather than a raw mismatch norm.

A.4 The angular reduction: exact form and four-term expansion

This appendix derives the two angular forms displayed in §3.4. Both follow from Eq. 1 and the split of δg by the native direction; no property of demotion enters. Write $g = \bar{g}_{\text{src}}$, $G = \|g\|$, $\hat{g} = g/G$, and $u^\perp = u - (\hat{g}^\top u)\,\hat{g}$; decompose $\delta g = G\,\kappa_\parallel\,\hat{g} + \delta g^\perp$ with $\kappa_\parallel = \hat{g}^\top \delta g / G$ and $\kappa_\perp = \|\delta g^\perp\|/G$.

The exact form. The two components are orthogonal, so $\|g + \delta g\|^2 = G^2(1+\kappa_\parallel)^2 + G^2\kappa_\perp^2$ and

$$\cos\big(g,\; g + \delta g\big) \;=\; \frac{\hat{g}^\top(g + \delta g)}{\|g + \delta g\|} \;=\; \frac{1 + \kappa_\parallel}{\sqrt{(1+\kappa_\parallel)^2 + \kappa_\perp^2}} \;=\; \frac{1}{\sqrt{1 + \kappa_{\text{eff}}^2}}, \qquad \kappa_{\text{eff}} = \frac{\kappa_\perp}{1 + \kappa_\parallel}, \tag{19}$$

valid whenever $1 + \kappa_\parallel > 0$ (the perturbation does not reverse the gradient direction). The gap of Eq. 1 is therefore exactly the saturating form displayed as Eq. 6 in §3.4. Since $1 - \cos(\hat{g}_1, \hat{g}_2) = \frac{1}{2}\|\hat{g}_1 - \hat{g}_2\|^2$ for unit vectors, starting from either expression in Eq. 1 lands on the same result.

The quadratic limit. For small perturbations, $1 - (1 + \kappa_{\text{eff}}^2)^{-1/2} \;=\; \frac{1}{2}\kappa_{\text{eff}}^2 + O(\kappa_{\text{eff}}^4)$ and $\kappa_{\text{eff}}^2 \;=\; \kappa_\perp^2\big(1 - 2\kappa_\parallel + O(\kappa_\parallel^2)\big)$, so

$$d_e \;=\; \frac{\|\delta g^\perp\|^2}{2G^2} \;+\; \underbrace{O(\kappa_\perp^2 \kappa_\parallel) + O(\kappa_\perp^4)}_{\text{beyond-quadratic}}. \tag{20}$$

The four-term expansion. Substituting $\delta g = B_e + C_e$ (Eq. 11) into the leading term and expanding the square,

$$\|\delta g^\perp\|^2 = \|B_e^\perp\|^2 + \|C_e^\perp\|^2 + 2\,\langle B_e^\perp,\, C_e^\perp\rangle,$$

exactly — the legs telescope, so no cross term with a remainder branch arises. Divided by $2G^2$, the three terms are the data share S_e, the graph share Φ_e, and the projected interaction I_e of the four-term expansion, Eq. 7. The remainder R'_e is therefore purely geometric, with an exact expansion: the beyond-quadratic terms of Eq. 20 and the parallel-component correction, each suppressed by an extra power of perturbation size relative to the retained terms in the small-κ domain.

From the four-term form to the excess. The reported object is the excess over the control. The reenc arm changes no grid, so its graph leg is identically zero, and both control terms built from it vanish: its graph share Φ_{reenc} and its projected interaction I_{reenc} each carry a factor of the vanished leg. Writing Eq. 7 for each arm and subtracting term by term,

$$\begin{aligned}
\text{gap}_e(\sigma) &= d_e(\sigma) - d_{\text{reenc}}(\sigma) \\
&\approx \big(S_e - S_{\text{reenc}}\big) + \big(\Phi_e - \Phi_{\text{reenc}}\big) + \big(I_e - I_{\text{reenc}}\big) + \big(R'_e - R'_{\text{reenc}}\big) && \text{Eq. 7, each arm} \\
&= \big(S_e - S_{\text{reenc}}\big) + \Phi_e + I_e + \big(R'_e - R'_{\text{reenc}}\big), && \Phi_{\text{reenc}} = I_{\text{reenc}} = 0
\end{aligned}$$

so the control collapses to $d_{\text{reenc}} \approx S_{\text{reenc}} + R'_{\text{reenc}}$, with S_{reenc} the re-encode share (measured ≈ 0 where probed: the control row of Table 8). The subtraction cancels only the re-encode component of the data share (up to a cross term suppressed by $\sqrt{S_{\text{reenc}}}$, bounded by the control's near-zero reading), so $\|\Delta\bar{r}\|$ is read net of re-encode; the graph share and the interaction pass through untouched, since the control has nothing to cancel them with.

The interaction bound, and the remainder's domain. The retained interaction obeys an exact Cauchy–Schwarz bound in the retained shares, and the remainder is a pure power of the perturbation amplitude:

$$|I_e| \le 2\sqrt{S_e\,\Phi_e}\,, \qquad \big|R'_e\big| = O\big(\kappa_\perp^2\kappa_\parallel,\ \kappa_\perp^4\big). \tag{21}$$

The first says the interaction can never exceed the retained sum ($2\sqrt{S_e\Phi_e} \le S_e + \Phi_e$) and would be automatically negligible wherever one share dominates the other; measured, the interference saturates this bound nearly everywhere (the near-cancellation of §4.3), which is why the main text carries it as entry (ii)'s measurement rather than an assumption. The second restates that the remainder is pure expansion geometry: it carries no branch-level content to bound, and its size is governed by the perturbation amplitude alone. Entry (i) of §3.4 handles it by domain rather than by assumption — at the measured in-window amplitudes ($\kappa_\perp \approx 0.5$–1.0) the truncation is out of domain, which is exactly the regime where the unit rule of §3.4 forbids quadratic magnitude reads and routes every magnitude through the exact link, under which nothing is truncated (the recorded ~3–4× underprediction of $h(B{+}C)$ by the truncated ledger, below, is this term made visible). The control counterpart R'_{reenc} needs no bound of its own: the control's degenerate expansion is itself the measured near-zero row of Table 8, which pins R'_{reenc} jointly with S_{reenc}.

What is not derived. The loading $\|\delta g^\perp\| = a_e\,\|\Delta\bar{r}(\sigma)\|$ (that the orthogonal gradient mismatch is proportional to the residual mismatch with a σ-independent route gain) is an empirical ingredient (the data-branch factorization at the amplitude level), fit and tested held-out in Appendix C. The geometry above fixes only how a given δg is read into gap units. What *is* derivable from first principles is only a one-sided envelope, $\|\bar{J}^\top\Delta\bar{r}\| \le \sigma_{\max}(\bar{J})\,\|\Delta\bar{r}(\sigma)\|$, which would turn the data term into an inequality in the observed mismatch norm given a σ-uniform bound on the mean Jacobian's top singular value. Promoting the envelope to a proportionality with a σ-independent gain requires the mismatch *direction* $\widehat{\Delta\bar{r}}(\sigma)$ to hold a σ-stationary alignment with $\bar{J}$'s singular directions (isotropy, for instance, would give gain $\|\bar{J}\|_F/\sqrt{n}$) while $\bar{J}$ itself varies with σ through the network input; the image- and route-specific mismatch directions of Appendix B license no such axiom. Hence the loading stays an assumption with a designated probe rather than a theorem.

B Instrument details and secondary reads

Estimator. Per image, arm, and σ-bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at $D{=}8$ are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is not used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in pre-debiasing runs its gap served as the validity band (± 0.04; observed $|\cdot| \le 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs), a role superseded by the paired debiased read of §4.1. The redraw floor plays the same pricing role for finite-draw noise.

Density-weighting the uniform bins by the trainer's σ-density reproduces earlier σ-marginalized measurements from the same instrument family (consistency check).

Debiasing, implementation. Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 9 is then computed per image and bin, and the map read is the paired excess $\mathrm{gap}_{e,i} = \widehat{\mathrm{gap}}_{e,i} - \widehat{\mathrm{gap}}_{\mathrm{reenc},i}$ with non-finite values (negative self-cosines under the square root at low D) and $|\mathrm{gap}_{e,i}| > 1.5$ trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ($D = 64$ contains the $D \leq 32$ sets; prefix sums, no extra forwards) and the c/D fit of §4.1 is run on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path sensitive: twin runs sharing a warm compiler kernel cache agree to $|\Delta \cos| \leq 0.015$, while a run compiled to a different kernel set lands up to ≈ 0.3 away at D=2, so gap/floor/self-floor pairings are not compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.

Finite-draw bias, measured. Detail behind the magnitudes quoted in §4.1: the c/D fit to the uncorrected 1024→896 endpoint decay gives $c \approx 0.5$. The native floor makes the point sharpest: at the verdict grid's D=8–16 the uncorrected endpoint self-cosine reads 0.79–0.85, a naive read pricing the native gradient's agreement with *itself* at a 0.15–0.2 gap, before the nested sweep ($D = 4 \ldots 64$) extrapolates it to the unit self-cosine reported in §4.1.

The U-shaped denominator (behind §4.4). The regimes of the total gradient norm $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ ($2.6 \to 0.4$ at $\sigma \approx 0.3 \to 9.3$; Fig. 13b): large at high σ, where the model pulls composition out of the prior; small in the mid-σ refinement regime; inflated again at low σ by irreducible ϵ. A monotone absolute mismatch over this U-shaped norm yields the measured mid-σ-peaked curves: bin-mean gap tracks $1/\|\bar{g}_{\mathrm{src}}\|$ (Spearman +0.55–+0.90). The renormalization check reads the amplification directly: multiplying each bin-mean gap by $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ flips the anomalous low-σ dip into the monotone-in-σ maximum in every arm of both verdict runs, which is the low-norm amplification behind the mid-σ peak of the measured curves. This shape read is post-hoc consistency analysis; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

Endpoint and x-zero modes. The σ=1 endpoint bin feeds input exactly ϵ (the input-mediated data term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low-σ bins are off-manifold and the σ=1 read is primary. In that regime the residual is $\approx -\hat{x}_{\mathrm{prior}}$, so "graph-dominated" includes the model's grid-conditioned prior; the prior-distance probe (§4.4), a forward-only comparison of the model's caption-conditioned prior across grids, subsequently dissociated the prior from the floor's route ordering: its route distances are flat where the floors are strongly ordered, ruling out a resolution-conditioned prior or a training-distribution discontinuity as the carrier. In the target-strength (α) sweep the mid-sweep points $\alpha \in [0.5, 0.75]$ are unreadable by construction and excluded: there the native residual $\alpha x - \hat{x}$ passes near cancellation, the gradient norm dips ($60 \to 33$) and the redraw floor falls to 0.87, so the estimator reads through a small gradient norm: the low-norm amplification above, surfacing along the α axis. The well-conditioned anchors behind the α-flatness claim of §4.4: 768 reads $+0.070 \pm 0.015$ at α=0 vs $+0.049 \pm 0.010$ at α=1 (control slope +0.003), and 512 reads $+0.269 \pm 0.041$ vs $+0.337 \pm 0.067$.

Parallel-landing decomposition of the target term (behind §4.4). The α-flatness has an exact mechanism. Because the objective is quadratic, $\bar{g}$ is affine in α, so the exact target-mediated gradient $t = \bar{g}(1) - \bar{g}(0)$ is computable from the endpoint arms; measured, it is real and large ($\|t_{\mathrm{src}}\|/\|\bar{g}_{\mathrm{src}}\| \approx 2.2$, so $J^{\top}$ does not annihilate target content), but demotion's change to it lands almost entirely *along* $\hat{g}_{\mathrm{src}}$ ($\kappa_{\parallel} = -0.75/-1.18/-1.86$ against $\kappa_{\perp} = 0.09/0.14/0.20$, an 8–9× parallel dominance reproducible across draw sets). The cosine gap is blind to a parallel rescaling (Eq. 6) and the orthogonal part enters only at second order ($\kappa_{\perp}^2/2 \approx 0.004$–$0.02$), so the α-flat result stands *because* the target term lands parallel, not because it vanishes. In this read the per-image orthogonal loading is ~10× the aggregate's: image-specific directions cancel in the mean, the same motif as the residual-direction read below. For 896 and 768 the floors sit at or below the verdict grid's resolution ε^*, which the endpoint sweeps resolve only through their order-of-magnitude larger draw budget.

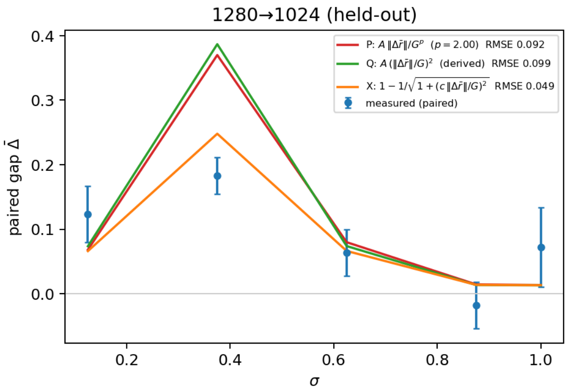


Figure 6: The second held-out route, 1280→1024, under all three functional forms for the data term of Appendix C (fitted power, derived quadratic, exact angular link; legend $G := \|\bar{g}_{\rm src}(\sigma)\|$), predicted from governors fitted on other routes. This is the one route where the forms visibly separate: the derived quadratic systematically overshoots the mid-σ peak; the exact angular link's saturation removes the overshoot (RMSE 0.049 vs $\approx$ 0.095), which is why the exact link is the form displayed in Fig. 2.

The statement "the high-σ plateau is the floor" therefore sharpens to: the floor is what the plateau converges to once the data term vanishes and the estimator can resolve it.

Direction-resolved mismatch (behind §4.4). The route-uniformity of $\|\Delta\bar{r}(\sigma)\|$ is a property of the *amplitude* only. Saving the per-image mismatch vectors and comparing $\Delta\bar{r}$ directions with a split-half attenuation correction, non-adjacent route pairs are near-orthogonal at low σ (corrected cosine 0.00–0.08 at $\sigma \le 0.375$) with only a weak shared component at high σ (+0.2–0.3 at $\sigma \ge 0.875$); the top mode of a stacked SVD carries 0.33–0.36 of the energy against the 0.25 rank-4-uniform baseline (a rank-one common mode would give ≈ 1); and cross-image direction consistency is zero at every (route, σ) cell (max +0.019 at split-half reliabilities 0.73–0.99), so the mismatch direction is fully image-specific, refuting a grid-conditional composition prior ("small canvas $\Rightarrow$ portrait framing") as the carrier, at least under full captions. What *is* shared is scale and spectral drift: $\Delta\bar{r}$ energy migrates toward low frequencies as σ rises (low-third share 0.40 $\longrightarrow$ 0.67 by $\sigma = 0.875$), composition-scale but per-image. Eq. 8 only ever consumes the norm, so the account is untouched; what sharpens is the open question, now "why is only the *amplitude* universal when the directions are image- and route-specific." A closed-form candidate for cross-band coupling exists, namely the flow posterior covariance, whose off-diagonal $D_z v^*$ entries are $-\big((1-\sigma)/\sigma^3\big)\,\mathrm{Cov}(x_\omega, x_{\omega'} \mid z)$ (Xing et al., 2026) and are set to zero by the diagonal-Gaussian premise; but its rank-one common-mode version is excluded by the SVD read above.

Held-out validation and refit, sources. The fit consumes the debiased paired curves of the 1024-tier verdict run, the 1280-tier curves (pre-debiasing paired reads; see §6), the route-uniform mismatch norm $\|\Delta\bar{r}(\sigma)\|$ from the residual probe, and each run's own bin-mean gradient norm $\|\bar{g}_{\rm src}(\sigma)\|$ (gap and norm always from the same process, per the kernel-path rule). The shared-power fit scans $p \in [1, 2]$ jointly across fit routes with per-route weighted least squares; the exact-link fit is a per-route grid with profile-likelihood uncertainties on a_e. Oracle baselines are quadratic-in-σ fits on the held-out data itself (the noise ceiling for a 3-dof curve); the spectral baseline on its committed region is gap = 0 for $\sigma > \sigma_{\rm eq}$, and its curve-level transport is the bridge of Fig. 11.

The floor law's identifiability. Two anchors and one held-out check cannot identify a functional form, and the cosine gap is the wrong scale to fit one in: it saturates (Eq. 6). Refitting the same anchors in the unsaturated perturbation-energy units $\kappa^2_{\rm eff} = (1 - F)^{-2} - 1$ gives $\tau \approx 860$ tokens and predicts $F(2160) = +0.096$, equally consistent with the measured $+0.092 \pm 0.012$; the same two anchors pushed through $e^{-\sqrt{n}/\ell}$ or a power law n^{-p} predict +0.085 and +0.075. Every form falls within the union of the measured CI and the prediction's own bootstrap spread, so the exponential is one member of a family our operating points cannot distinguish, and the mechanical reading of τ ($\approx$ 860–1040 tokens across the two unit conventions) as an effective-rank decay length of the early-block gradient operator is a hypothesis, not a law. It is a discriminable one: a spectral-tail account predicts a source-capacity dependence $F \propto e^{-n/\tau} - e^{-n_0/\tau}$ where the absolute-target-capacity governor predicts none (at our operating points the secant correction sits within calibration slack, so the existing anchors do not discriminate); the designated run is a fixed-target, varied-source token ladder, not yet scheduled.

Bootstrap bands and the leave-896-out lane (Fig. 2). The bands are a full-pipeline image bootstrap (B=1000, fixed seed; 981 draws kept): resample images with replacement within each run (N=40 / N=24), re-bin the paired curves, refit the exact-link (a_e, Φ_e) on every fit route, re-derive the ratio governor and floor law, re-predict. $\|\Delta\bar{r}(\sigma)\|$ and the bin-mean $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ are run-level instruments with no per-image decomposition and are held fixed. The replica includes the committed floor law's positive-floor filter ($\Phi_e > 0.005$): in 41% of draws the resampled 1120 floor clears the filter and the law re-anchors on the $(896, 1120)$ legs rather than $(512, 896)$. The bands therefore contain this branch variability, which is the reason §5 deploys the measured map rather than the predicted one. The 896 curve in Fig. 2 is the post-hoc leave-896-out lane under the exact link (c from the ratio-twin 1120 alone = 0.142 vs 0.150 fitted on 896 itself; floor law through 512+1120, whose fragile Φ_{1120} leg enters the log fit clamped positive (36% of bootstrap draws clamp), giving $F(3012)$ = +0.012 vs measured +0.019 [+0.010, +0.030]; curve RMSE 0.073).

Interventional B/C ledger, implementation. Routes 1024→{896, 768, 512}, $\sigma \in [0.5, 1.0]$ in four bins plus the σ=1 endpoint bin, D=8 draws per bin, N=24 images, deterministic mode, two independent draw sets per arm. Each arm's draw-summed adapter gradient is stored per bin; the interventional split realizes the legs of Eq. 5: $B = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{src}}$ (the data intervention at fixed native graph: the demote–re-promote arm's input passed through the downscale→upscale→encode pipeline but evaluated on the native grid) and $C = \bar{g}_{\mathrm{demote}} - \bar{g}_{\mathrm{repromote}}$ (the graph intervention at fixed demoted data). The quadratic shares S, F, I and $\rho = \cos(B^\perp, C^\perp)$ use cross-draw-set inner products to cancel shared draw noise; the debiased estimates are not confined to $[-1, 1]$ (hence $\rho = -1.24$ at one low-amplitude bin). The exact counterfactual angles $h(B)$, $h(C)$, $h(B{+}C)$, with $h(X) = 1 - \cos(\bar{g}_{\mathrm{src}}, \bar{g}_{\mathrm{src}} + X)$, are computed on the raw arm means; since $\bar{g}_{\mathrm{dem}} = \bar{g}_{\mathrm{src}} + B + C$ by construction, the realized demotion distance is exactly $h(B{+}C)$, so each account's predicted-to-realized ratio is direct. The no-interference scalar account $S{+}F$ overpredicts the realized in-window gap 2.4–3.8× at 768 (1.6–5.8× at 896, 1.2–3.0× at 512); the fully additive counterfactual $h(B) + h(C)$ overpredicts 3.5–8×, the realized $h(B{+}C)$ being only ∼20–30% of the additive sum in-window. Including I is what removes the overprediction; the vector account $B{+}C$ is exact by construction. One caveat on the units of Table 3: in-window $|B^\perp|, |C^\perp| \approx 0.5$–$1.0\,\|\bar{g}_{\mathrm{src}}\|$, outside the quadratic truncation's domain, so the truncated ledger $S{+}F{+}I$ *under*predicts realized magnitudes ∼3–4× (median ratio 0.24×). The ledger licenses sign, decomposition, and window localization; gap magnitudes are quoted from $h(\cdot)$. Finally, the re-encoding proxy check re-references the data leg against the control arm, $B_{\mathrm{reenc}} = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{reenc}}$: $|B^\perp_{\mathrm{reenc}}|$ agrees with $|B^\perp|$ to ∼4% at the signal-carrying low-σ bins (worst ±23% only where B is already small), so the shared downscale→encode pipeline cost is a minor share of the data intervention. This closes the one open item on the floor ledger's first share, where the re-encoding control (native decode→re-encode) is a *proxy* for the pipeline cost demotion actually pays.

The ledger's geometry: why the legs anti-align. The near anti-parallelism of Table 3 is structural rather than incidental (Fig. 7). The three arm means are values of one map on a 2×2 data–graph lattice — $\bar{g}_{\mathrm{src}}$ at (native content, fine graph), $\bar{g}_{\mathrm{repromote}}$ at (demoted content, fine graph), $\bar{g}_{\mathrm{dem}}$ at (demoted content, coarse graph) — all living in the shared adapter-parameter space, so the differences are literal, and B and C are the partial differences along the path (native, fine) → (demoted, fine) → (demoted, coarse). That is the lattice's only realizable path: its fourth corner, native content on the coarse graph, does not exist, since placing native content on the coarse grid *is* the demotion pipeline — the coarse graph forces the demoted data. The endpoints are the two content–graph *matched* corners (native content on its own grid; band-limited content on the grid whose Nyquist matches it); the unavoidable waypoint is the single mismatched corner. The split is therefore a round trip off the matched configuration and back — B creates the content–graph mismatch at fixed graph, C removes it at fixed data — so, to leading order in the mismatch, whatever share of each leg passes through it is the same vector with opposite sign, which fixes the sign of ρ. Its depth is then bookkeeping: projection off $\hat{g}_{\mathrm{src}}$ is linear, so $|(B{+}C)^\perp|^2 = |B^\perp|^2 + |C^\perp|^2 + 2\rho\,|B^\perp||C^\perp|$, and the ledger carries

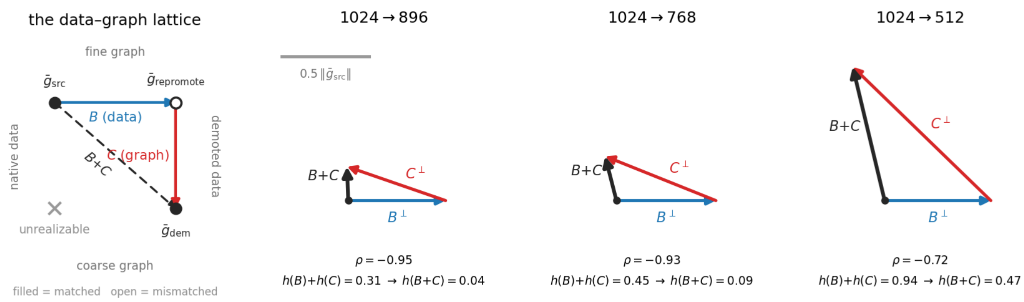


Figure 7: **The ledger's geometry.** Left: the three arms as corners of the data–graph lattice. B and C are the partial differences along the lattice's only realizable path — the fourth corner (native content on the coarse graph) does not exist — so the split is a round trip off the matched configuration ($\bar{g}_{\mathrm{src}}$, $\bar{g}_{\mathrm{dem}}$) through the single mismatched corner ($\bar{g}_{\mathrm{repromote}}$). Right: the measured legs tip-to-tail at $\sigma = 0.563$ (Table 3), to a common scale in units of $\|\bar{g}_{\mathrm{src}}\|$; the drawn angle is the debiased ρ. Amplitude-matched legs leave only a small resultant (896, 768); the 512 graph leg is ~1.8× the data leg, and the unmatched share survives as the realized gap.

two independent measurements — how far the waypoint sits from the corners (the leg amplitudes) and how nearly the endpoints coincide (the resultant) — with ρ their law-of-cosines consequence; the separately debiased sign column is a consistency check on the raw $h(\cdot)$ reads, not a third fact. Two corollaries follow, and the data confirms both. Route verdicts reduce to amplitude matching, the realized $h(B{+}C)$ collapsing where $|B^\perp| \approx |C^\perp|$ (the window-center localization of Appendix C); and what no matching can cancel is the graph-intrinsic share of C with no data-side counterpart — the resultant collects exactly the residue the floor ledger below decomposes into $\mathrm{RoPE}_e + \mathrm{Resid}_e$, and at the $\sigma = 1$ endpoint, where the input is ϵ on every arm and the data leg has nothing left to act on, that uncancelled residue is the whole of the floor.

Floor decomposition, detail (behind §4.4). Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) localizes the floor in *depth*, not module type: early blocks (0–9, peaking at 3–8) carry ~3× the late-block gap, uniformly across every module type within a block (at $512/\sigma{=}1$ every type sits in 0.22–0.31; Table 13). The mechanism picture: the first ~10 blocks build grid-calibrated token statistics, the divergence propagates into every parameter type's gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Query and key projections show *zero* excess over value projections, which we initially read as refuting a rotary mechanism. That was an instructive error, since landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. Hence the origin-side positional-interpolation intervention (Table 14; the paired Δs are same-grid contrasts at matched draws, in which the finite-draw bias cancels to first order). The 896 route's small floor sits below that probe's paired resolution and remains undecomposed. A σ-resolved follow-up forecloses the route this suggests (training 1024⟶768 through interpolated coordinates): with image content in the input the stretched forward is off-manifold, and the interpolated arm is *worse* than the plain demoted arm through $\sigma \in [0.56, 0.81]$ (paired −0.05 to −0.11), winning only at $\sigma \geq 0.94$; the 768 route's data term is in any case fatal on its own (+0.09–0.22 over control across the window). On the non-positional residue, the attention-over-N piece has inference-side prior art: attention-entropy corrections derive length-dependent temperatures to preserve the $\log N$ entropy term (Li et al., 2025), and TIDE applies the same dilution correction jointly to resolution and diffusion timestep (Liu et al., 2026), the closest prior work to a σ-gated resolution correction, though on the inference side; our gate is training-side and set by measured gradient safety, not by an entropy invariance.

The floor as a ledger. Combining the designated probes of §4.4, the floor decomposes additively, each share measured by its own intervention:

$$\Phi_e \;=\; \underbrace{\text{reenc}}_{\approx 0 \text{ by control}} \;+\; \underbrace{\text{target content}}_{\text{lands } \parallel\, \hat{g}_{\mathrm{src}}\text{: angular share}\,\approx\, 0} \;+\; \underbrace{\mathrm{RoPE}_e}_{\text{erased by PI at } \sigma=1} \;+\; \underbrace{\mathrm{Resid}_e}_{\text{remainder}}$$

Table 3: The interventional B/C ledger behind Appendix C: per (route, σ-bin) orthogonal amplitudes (units of $\|\bar{g}_{\mathrm{src}}\|$), interaction geometry, the cross-set-debiased quadratic shares, and the exact counterfactual angles. In-window S+F+I is sign- and localization-accurate but underpredicts $h(B{+}C)$ (small-perturbation truncation out of domain); magnitude claims in the text use $h(\cdot)$.

route	σ	$\lvert B^{\perp}\rvert$	$\lvert C^{\perp}\rvert$	ρ	S	F	I	$h(B)$	$h(C)$	$h(B{+}C)$
896	0.563	0.55	0.59	−0.95	0.103	0.115	−0.206	0.107	0.206	0.037
	0.688	0.42	0.39	−0.91	0.036	0.059	−0.085	0.083	0.077	0.062
	0.813	0.20	0.23	−0.94	0.016	0.019	−0.033	0.022	0.027	0.010
	0.938	0.10	0.10	−1.24	0.002	0.003	−0.006	0.007	0.006	0.003
	1.000	0.05	0.06	−0.71	0.001	0.001	−0.001	0.002	0.003	0.002
768	0.563	0.57	0.68	−0.93	0.139	0.200	−0.309	0.107	0.346	0.088
	0.688	0.56	0.57	−0.93	0.100	0.136	−0.218	0.106	0.242	0.098
	0.813	0.35	0.42	−0.93	0.049	0.067	−0.105	0.059	0.110	0.035
	0.938	0.14	0.17	−1.02	0.005	0.009	−0.014	0.010	0.033	0.008
	1.000	0.07	0.09	−0.62	0.001	0.004	−0.003	0.003	0.014	0.013
512	0.563	0.60	1.09	−0.72	0.166	0.403	−0.372	0.085	0.854	0.466
	0.688	0.63	0.88	−0.86	0.153	0.332	−0.388	0.107	0.623	0.305
	0.813	0.57	0.72	−0.87	0.152	0.226	−0.323	0.131	0.285	0.126
	0.938	0.25	0.29	−0.96	0.026	0.036	−0.058	0.028	0.201	0.037
	1.000	0.13	0.15	−0.88	0.007	0.010	−0.015	0.007	0.379	0.098

Table 4: **Head to head** on the same measured curves: RMSE in cosine units over all σ-bins. The spectral column is the family transported through our bridge (Fig. 11) and is δ-inert. Our two columns are the derived small-perturbation form (Eq. 8) and the exact angular link (Eq. 6), both with *zero per-route freedom* on held-out routes: amplitudes and floors come from the two governors of §4.4 fitted on {1024→896, 1024→512, 1280→1120}. The oracle is a quadratic-in-σ fit on the held-out data itself, the noise ceiling for a 3-dof curve, so beating it on 1024→768 means the prediction is at the data's own resolution.

route	status	spectral	two-term	exact link	oracle
1024→768	held out	0.147	0.093	0.093	0.105
1280→1024	held out	—	0.092	**0.049**	0.056
1024→512	fit route	0.355	0.093	—	—

with totals 0.02–0.04 (896; below the PI probe's paired resolution, undecomposed), ≈ 0.07 (768; RoPE the large majority, Resid ≈ 0), and ≈ 0.30 (512; RoPE ≈ 0.10, Resid ≈ 0.20).

C Head to head: predicting held-out routes

The probes of §4.4 test each account's *structure*. The remaining test is predictive: fit the two-term reduction on three routes, then predict two held-out routes from the governors alone. Fit routes: {1024→896, 1024→512, 1280→1120}; held out: {1024→768, 1280→1024}. Per fit route the excess curve $\overline{\mathrm{gap}}_e(\sigma)$ is fit with a route amplitude and floor on the measured route-uniform $\|\Delta\bar{r}(\sigma)\|$ and each run's own $\|\bar{g}_{\mathrm{src}}(\sigma)\|$; two governor models ($A$(ratio) interpolating the fitted amplitudes, and an exponential floor law $F(n) = F_0 e^{-n/\tau}$ in target tokens n) then generate the held-out predictions with zero per-route freedom. Pass criteria were pre-registered in the analysis script before the numbers.

Held-out results, and the comparison. All four pre-registered gates pass (Table 4, Fig. 2). (i) Held-out 1024→768: RMSE 0.093, *better than an oracle quadratic fit on the held-out data itself* (0.105), and better than the spectral account both across all bins (0.147) and on the spectral account's own committed region, where it predicts gap 0 (0.164 against 0.096 for ours). (ii) Held-out 1280→1024: RMSE 0.092 under the derived quadratic and 0.049 under the exact link, against oracle 0.056; inside the pre-registered 2× gate either way. (iii) The ratio governor holds at the amplitude level: the two ratio-0.875 routes agree at $z = 0.14$ despite a 1.6× difference in target capacity. (iv) The floor law $F(n) = 0.70\, e^{-n/1041\,\mathrm{tok}}$, fit on the 512 and 896 floors, predicts $F(4825) = +0.007$ for the 1120 grid (fitted +0.002) and $F(2160) = +0.088$ for the held-out 768 grid, against a measured endpoint floor of $+0.092 \pm 0.012$; a post-hoc leave-896-out re-derivation from 512+1120 alone predicts the 896 floor at +0.019 under the fitted-power form (+0.012 under the exact link, Appendix B) against the measured +0.019 $[+0.010, +0.030]$. The floor law is thus correct on both floors outside its fit set, though its point estimates vary with the draw: a full-pipeline bootstrap puts $F(2160)$ at 68% $[+0.042, +0.091]$, so the licensed read is agreement, not exactness.

The margin in Table 4 has a structural reading, not merely a numerical one. The spectral family's error is not a bad tolerance, since δ is inert, but a missing term and a missing governor: it has nothing to put in the high-σ region where the floor lives, and nothing to distinguish 1280→1024 from a same-ratio route at a quarter of the token count. Our account's held-out accuracy comes from precisely the two ingredients it adds.

Which functional form, and what the fit does not identify. Refitting the same pipeline under three forms for the data term (a fitted power $A\, m/G^p$ with p shared and free, the derived quadratic, and the exact link) resolves the form question in the geometry's favor twice over. The free exponent's own optimum lands on $p = 2.00$ exactly, recovering the derived quadratic on its own; and the exact link is the best held-out predictor (mean RMSE 0.071 vs 0.093–0.094), with its entire margin coming from 1280→1024, where the quadratic systematically *overshoots* the mid-σ peak and the exact link's saturation removes the overshoot (Appendix Fig. 6); the fitted governors themselves come out form-invariant. Two things the fit does *not* establish: the prediction is not within the verdict resolution anywhere (χ^2/bin 7.6–9.2 held-out under the quadratic), so the licensed claim is shape, magnitude class, and governors at ~0.07–0.09 RMSE; and the floor law's functional form is unidentified at our operating points (the identifiability analysis is in Appendix B).

The reduction's domain: one signed failure, and its mechanism. The 768 route's mid-σ window is the reduction's one structural failure, and its shape is diagnostic. Measured excess sits at zero across $\sigma \in [0.56, 0.94]$, *below* the predicted $\Phi_{768} \approx 0.09$ and outside the bootstrap 95% band (Fig. 2): a positive two-term reduction cannot produce a gap under its own floor, under any coefficients. In the four-term expansion (Eq. 7) this admits exactly two mechanisms, both pre-registered ahead of the probe: either the projected interaction is negative in the window ($I_{768} < 0$, in which case amplitude matching predicts the window center sits where the two perturbation legs have equal orthogonal magnitude, a testable localization), or the graph share is itself σ-dependent, rewriting assumption (iii) rather than entry (ii). The designated probe is vector-resolved: a demote–re-promote arm reads the shares per (route, bin) from the interventional split $B = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{src}}$, $C = \bar{g}_{\mathrm{demote}} - \bar{g}_{\mathrm{repromote}}$ (Appendix B, Table 3).

When the probe is run, the data select the first branch, with its localization confirmed: $I_{768}(\sigma) < 0$ at every bin (-0.31 at $\sigma = 0.56$, shrinking to -0.014 by 0.94), with the two legs near anti-parallel in-window ($\rho = \cos(B^{\perp}, C^{\perp}) \approx -0.93$), and the predicted window center lands at $\sigma \approx 0.69$ ($|B^{\perp}|/|C^{\perp}| = 0.98$ there), inside the measured window. Branch (ii) is ruled out: $\Phi_{768}(\sigma)$ decreases monotonically to its endpoint value and never drops below it, so the endpoint reads of §4.4 are untouched. The anti-parallel geometry turns out to be universal ($\rho \in [-1.24, -0.62]$ across every route and bin), as the round-trip reading of the split says it must be (Appendix B, *the ledger's geometry*), so what distinguishes routes is amplitude matching, not sign; measured against the exact counterfactual angles, roughly three quarters of the additively-predicted gap is erased by the interference in-window (Appendix B). The window therefore *delimits the reduction's domain*, and the derived four-term form represents the failure rather than excusing it.

D The measured geometry: supporting panels

The two figures above carry the supporting detail for §4.3: the lane test of the cancellation-aware form, and the reliability and frame-transport reads that scope the axis-field claims.

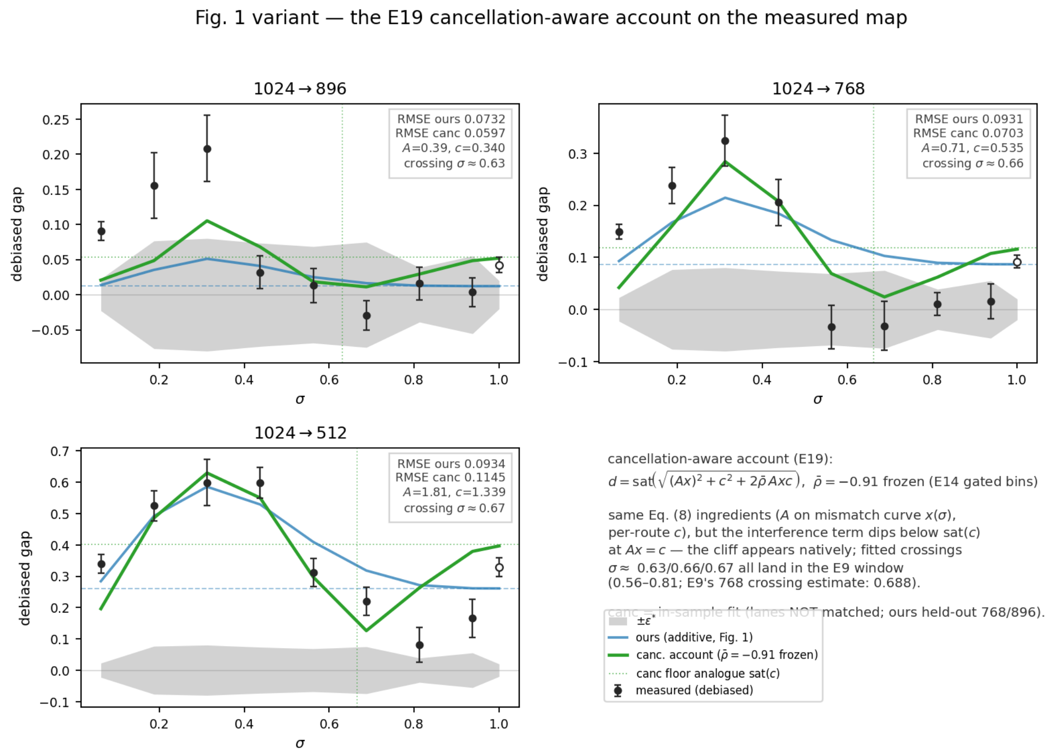
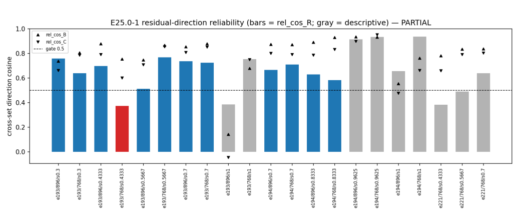
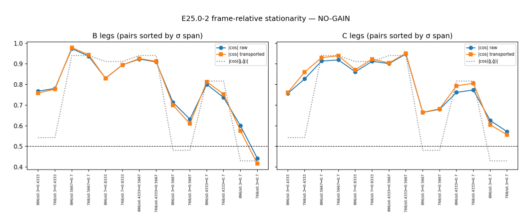


Figure 8: **The cancellation-aware operational form (demonstration figure; in-sample).** The exact link (Eq. 6) evaluated on *both* shares with the interference at the pooled constant $\bar{\rho} = -0.91$, drawn over the measured curves: in-sample it reproduces the map's shape including the 768 in-window dip below the route's own floor, the structure the additive two-term form cannot express. This panel is a demonstration, not lane evidence; the lane-matched score (§4.4, identical protocol with only the link swapped) is: held-out 768 RMSE 0.078 vs. additive 0.099 with the dip placed at $\sigma = 0.67$ from held-out parameters alone; leave-one-out 896 *lost* at 0.086 vs. 0.077; 1280-tier guard broken on 1024 held-out (0.199 vs. 0.046). Reported at lane parity; not adopted — the additive lane keeps Fig. 2.



(a) Reliability of the pooled residual direction $\widehat{R} = \widehat{(B{+}C)}$: cross-draw-set cosine per verdict condition against the 0.5 gate. 11 of 12 conditions pass; the one failure is $768/\sigma = 0.43$, and mid-window 768 is where the direction is least reproducible.



(b) Matched-angle, not planar: transporting a bin's pooled leg direction by the native anchor $\hat{g}(\sigma)$'s own measured rotation before comparing across σ changes the cosines by a median of -0.0005 — the axis rotates through the same angle as the anchor but in its own plane, so a lookup in an anchor-normalized frame is refuted.

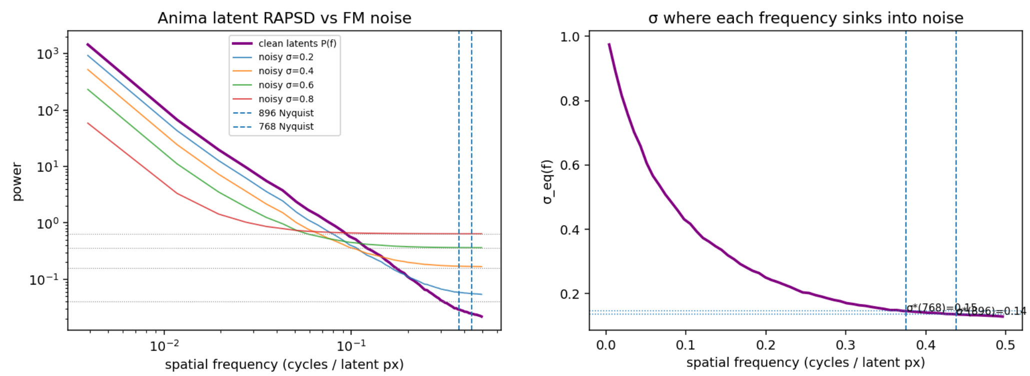
Figure 9: Supporting panels for the axis-field reads of §4.3 (one adapter, three vector stores, standard corpus; pooled directions only).

E The spectral account: spectrum and instantiation

The tolerance δ of Eq. 4 is the account's one free parameter, and this appendix instantiates the whole family. The choice $\delta = (1 + P_\omega)/2$ recovers the often-quoted equal-power (input-SNR) crossover

Table 5: The spectral account (Eq. 4) instantiated on the measured latent spectrum (P at the 896/768/512 cuts = 0.025/0.029/0.065; RAPSD in Fig. 10): predicted safe boundaries t^* per route as the tolerance δ sweeps its range. δ_{reenc} is the measured re-encode noise floor, a zero-free-parameter anchor fixed by our own pipeline. No row reproduces the measured pattern (last line): the δ that matches the safe route's boundary predicts both failing routes safe by $\sigma = 0.63$, and no δ can produce a floor.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	default of Xiao et al. (2026)
$\delta_{\text{reenc}} \approx 10^{-5}$	0.98	0.98	0.99	the parameter-free anchor
measured, debiased	≈ 0.5	none safe	unsafe (8/9 bins)	floors .02–.04 / .06–.09 / ≈ .3



(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids' Nyquist cuts.

(b) The equal-power slice $\sigma_{\text{eq}}(f)$ of the spectral family. Predicted σ^*: 0.136 / 0.146 / ~0.20 for demotion to 896 / 768 / 512.

Figure 10: The spectral account, computed from the measured RAPSD and transported to the training question (§3.2): safety would be predicted from $\sigma \approx 0.14$. Against the measurement (Fig. 13) no tolerance reproduces the pattern (Table 5): the equal-power crossover is off by ~3.5×, and the 2× downscale, predicted safe by every tolerance, is safe at no noise level.

$\sigma_{\text{eq}} = \sqrt{P_\omega}/(1 + \sqrt{P_\omega})$ as one slice; Xiao et al. (2026) provide an inference-side default $\delta = 0.01$; and our own pipeline fixes a zero-free-parameter anchor δ_{reenc} by construction (below).

Table 5 evaluates Eq. 4 at the demoted grids' Nyquist frequencies $f_{\text{cut}} = \frac{1}{2}\, e/e_0$ on the N=40 mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065; Fig. 10). Across $\delta \in [0.001, 0.5]$ the mildest–harshest spread remains below 0.13, and the 1024→896 versus 896→768 boundaries (cut frequencies 0.4375 vs 0.4286) separate by no more than 0.01. The parameter-free anchor δ_{reenc} of §4.4 is measured by the same estimator: RAPSD of the re-encode arm's latent error (fresh resize–encode minus cached latent, the gradient probe's own control chain) on the same probe set, interpolated at each route's cut frequency. Measured (N=40, 64 radial bins): error variance 2.7×10^{-5} against latent variance 0.419; $D(f_{\text{cut}}) \approx 1.0 \times 10^{-5}$ at all three cuts (p10–p90 over images within $[1, 2] \times 10^{-5}$; per-band SNR $P(f_{\text{cut}})/D(f_{\text{cut}}) \approx 2.5$–$4.4 \times 10^3$), giving $t^*_{\text{reenc}} = 0.98/0.98/0.99$ for 896/768/512.

The "wrong δ" objection, closed by construction. One might object that the right δ has simply not been chosen: SPD's is an inference-side value selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026), and nothing ties it to gradients. Our setting, however, fixes one *by construction*: the demotion recipe pays a resize–VAE-encode round trip whose latent error has measurable per-band power $D(f)$, expressed in Eq. 4's own units, so $\delta_{\text{reenc}}(e) := D(f_{\text{cut}}(e))$ anchors the family with zero free parameters. Measured (above), that anchor is tiny and image-generic, and the boundary moves it forces are measured in Table 5 and Fig. 11: every boundary is

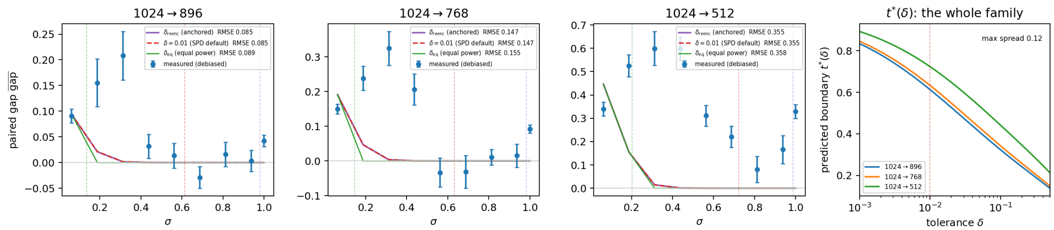
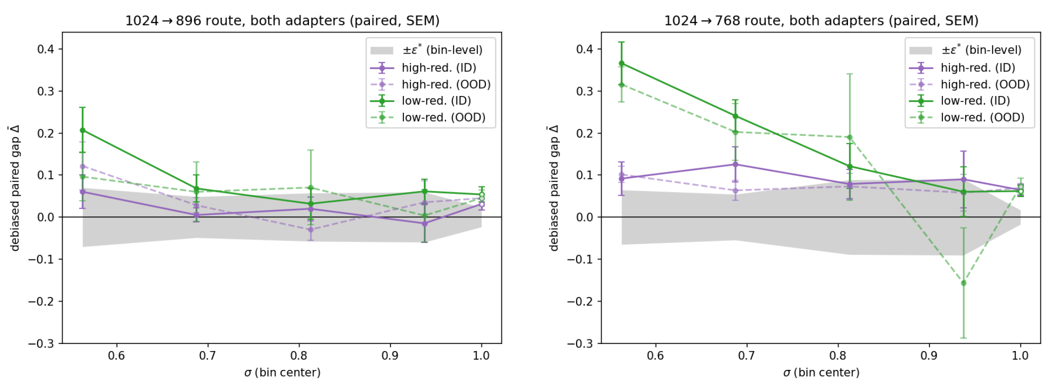


Figure 11: The *whole* tolerance family against the measurement: the detail behind the single red curve of Fig. 2. Left three panels: the measured debiased gap curves against the spectral account transported through our bridge (the diagonal model's destroyed-band mean-residual mismatch on the measured spectrum, through the measured $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ with a single gain calibrated on the one safe route and no floor, which the family cannot express), gated at $t^*(\delta)$ (dashed verticals) for the equal-power slice, SPD's default $\delta = 0.01$, and the measured anchor δ_{reenc}. The transported curve decays to zero by $\sigma \approx 0.35$ on every route, so the two larger gates (0.61–0.72 and 0.98–0.99) act on an already-zero curve and coincide exactly; the equal-power gate (0.14–0.20) is the only one that truncates a nonzero curve, and it moves RMSE by at most 0.008. All three therefore land within 0.01 of each other: δ is inert at the curve level. Right: the boundary family $t^*(\delta)$; maximum route spread 0.125 in σ over $\delta \in [10^{-3}, 0.5]$.



(a) 1024⟶896 (the safe route).

(b) 1024⟶768.

Figure 12: The Fig. 13a recipe repeated on two controlled adapters trained on opposite style clusters, one panel per route (N=48 stems per adapter, D=8/bin, high-σ window; same paired debiased estimator and trim; open markers the σ=1 endpoint bin; axes shared across panels). Unlike the main-text figures, color encodes the *adapter* (high- vs low-redundancy cluster); solid curves are in-distribution stems, the adapter's own style cluster (n=36: trained-on, held-out, and unseen-artist cells), and dashed curves are out-of-distribution stems from the opposite cluster (n=12). Gray band: bin-level $\pm\varepsilon^*$ per Eq. 10, median of the two runs' full-N SEMs for the panel's route. The map replicates: on the safe route all four adapter×distribution curves reach the band inside the window, on 768 all four sit above it until the top, and within each panel the dashed OOD curves track their solid ID counterparts.

pushed to $t^* \approx 0.98$, maximally conservative, and still structurally wrong in both directions at once: it moves all three boundaries together (spread < 0.01), predicts the measured-safe route safe only above $\sigma \approx 0.98$ where the measurement shows it safe from 0.5, and, like every member of the family, predicts the 2× downscale eventually safe when no noise level is. The confrontation of §4.4 therefore does not hinge on the choice of tolerance: the structural failures of §3.2(a–c) are δ-independent at the boundary level, and δ is inert outright at the curve level (Fig. 11).

F Controlled-adapter replication of the verdict map

The one-adapter limitation of §6 pre-registered a controlled 2×2 factorial: two plain-LoRA checkpoints trained under the original probe adapter's frozen recipe on opposite style clusters (the top and bottom 12 artists by median latent redundancy, hereafter the high-redundancy, visually flat, and low-redundancy, heavily textured, clusters), with stem-level membership manifests frozen before training (124 training images each, identical seeds). Each adapter was probed on 48 stems spanning four membership cells (N=12 each): trained-on, held-out images of trained artists, unseen artists of the same cluster, and unseen artists of the opposite cluster, with the cross-cluster stems shared between the two runs so cross-adapter contrasts read paired per stem. Figure 12 repeats the verdict-map recipe per route, overlaying both adapters.

Three reads, stated at the bin-mean level (per-cell tables reproduce from the archived per-image rows). First, the (route, σ) shape replicates on both checkpoints (Fig. 12). Second, the factorial contrasts are null at instrument resolution: the probe-style main effect and the adapter×probe-style interaction (the pre-registered verdict quantity) are both bounded below the resolvable bin-mean effect (raw-gap paired interaction -0.022 ± 0.027 at 896, -0.015 ± 0.059 at 768), and the membership contrast (trained-on versus held-out) does not replicate in sign across the two adapters. The map is therefore adapter- and content-agnostic on this model at our resolution, supporting a calibrate-once-per-model reading. Third, the one quantity that is *not* checkpoint-invariant is the absolute redraw-floor level: in-window floor cosines sit at ≈0.73 for the high-redundancy-cluster adapter versus ≈0.50 for the low-redundancy-cluster adapter, uniformly across all four membership cells. The floor's level is a property of the checkpoint (its gradient stochasticity), while the safety map built on top of it is not.

Cancellation replication on the controlled adapters. The same two checkpoints carry the cross-adapter replication of the cancellation geometry (§4.3, §6). The vector-resolved probe of §4.3 was re-run per adapter on the 1024→768 route over the five window bins (N=40, D=12, identical probe list; the operating adapter's reference rows from a same-environment twin run), with criteria frozen before any adapter gradient existed: a bin is *readable* iff both legs pass the reliability gate (rel ≥ 0.5), and an adapter *replicates* iff at least 4 of 5 bins are readable and every readable bin has $I < 0$, $\rho \leq -0.5$, and $h(B{+}C) < \min(h(B), h(C))$. Both adapters pass at 5 of 5 bins (Table 6). One consistency read is pre-registered with the table: from the measured h-triple the law of cosines already implies a ρ; at 9 of 10 adapter-bins the implied value is itself ≤ -0.97, so the deep measured ρ is there an arithmetic consequence of the magnitudes and is reported as such — the depth of enforcement is *not* claimed to scale with the perturbation. The single cell where measured enforcement exceeds what the magnitudes force (dirty, $\sigma = 0.7$: implied -0.87, measured -0.96) is recorded, not generalized. 896-route replication was not run and remains an open cell.

Table 6: Cross-adapter cancellation replication (1024→768, five window bins, frozen criteria; reenc-referenced $h(\cdot)$ units). Every bin of both adapters is readable and passes ($I < 0$, $\rho \leq -0.5$, $h(B{+}C) < \min(h(B), h(C))$): the realized gap is the residual of the same near-cancellation on adapters trained on disjoint data.

	flat-cluster adapter				textured-cluster adapter			
σ	$h(B)$	$h(C)$	$h(B{+}C)$	ρ	$h(B)$	$h(C)$	$h(B{+}C)$	ρ
0.30	.441	.337	.152	−.890	.647	.392	.233	−.923
0.43	.304	.477	.196	−.909	.453	.878	.420	−.936
0.57	.188	.357	.098	−.912	.451	.826	.400	−.948
0.70	.118	.221	.045	−.928	.532	.664	.333	−.964
0.83	.113	.108	.020	−.948	.820	.460	.214	−.980

G Render prompt for the arms figure

The renders of Fig. 5 share one positive prompt, a held-out corpus-B caption, rendered at 1024^2 with matched noise seed across arms:

```
safe, 2girls, kaai yuki, otomachi una, vocaloid, zako (vocaloid), @channel
(caststation), bag, black hair, black skirt, blush stickers, bow, bowtie, collared
```

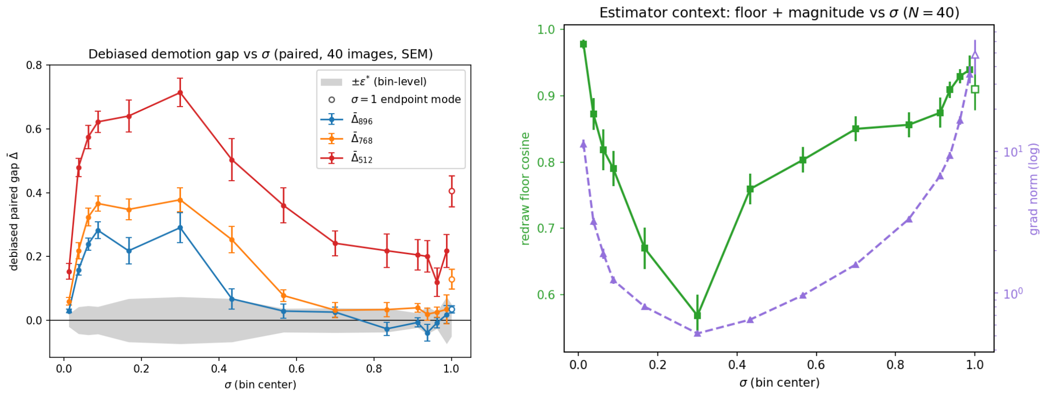


Figure 13: The measurement in detail. (a) Debiased demotion gap per σ-bin, all three routes on one shared axis, paired against the re-encoding control (1024-tier natives, N=40; per-bin values and SEMs in Table 7; gray band: bin-level $\pm\varepsilon^*$ (§4.1); open markers: the σ=1 endpoint bin): safety is a per-route boundary. The 896 route enters the band at $\sigma \approx 0.5$, 768 falls only to a low-excess shelf (+0.02–0.04 over $0.65 < \sigma < 0.95$, short of the strict margin), and 512 never falls inside the band (closest approach $+0.12 \pm 0.05$ at $\sigma = 0.96$); all three boundaries stand against the spectral family's prediction of $\sigma \approx 0.14$–0.20. (b) The estimator context those curves are read through: the redraw-floor cosine and the U-shaped gradient norm $\|\bar{g}_{\mathrm{src}}(\sigma)\|$.

```
shirt, diagonal-striped bow, diagonal-striped bowtie, diagonal-striped clothes,
diagonal-striped neckerchief, emphasis lines, hair bobbles, hair ornament, hands on
own cheeks, hands on own face, holding, holding phone, jitome, long sleeves, multiple
girls, multiple views, neckerchief, notice lines, one eye closed, outstretched arm,
phone, pink background, pink jacket, plaid clothes, plaid skirt, school bag, small
sweatdrop, striped bow, striped bowtie, striped clothes, striped neckerchief, tongue,
tongue out, v, v over eye, v over mouth, white shirt, wink star
```

H Per-bin verdict numbers, and the historical record

Table 7 is the numerical form of Fig. 13a, the paper's verdict map, and Table 8 the debiased floor reads that §4.4 scores the graph term on. The remaining tables are the raw-estimator record: retained for continuity and for the reads defended in §4.1 (one-signed bias; paired same-grid contrasts), not cited as debiased evidence. Table 9 is the uncorrected counterpart of the verdict map on the original coarse grid, and its last two rows are the coarse-grid estimator context (Fig. 13b plots the same two curves on the dense verdict grid).

Table 7: **Debiased verdict map** (plotted in Fig. 13a): paired per-image debiased excess $\overline{\text{gap}}$ (SEM) against the re-encoding control, 1024-tier natives (N=40, D=12/bin, deterministic kernels, on a segmented grid dense below $\sigma = 0.1$ and above 0.9: 14 bins in (0, 1) plus a σ=1 endpoint bin; trimmed per §4.1, retaining 36–40 images per cell). Bin-level ε^* at this (N, D) is 0.02–0.07. A reconstruction check (the realized excess rebuilt from per-arm means) flags the three $\sigma \leq 0.06$ bins as partly estimator inflation, so the raw-paired excess is the primary read there (896: +.037/+.118/+.221; 768: +.065/+.202/+.275; 512: +.172/+.406/+.497), up to 0.08 smaller with the same verdicts.

σ bin center	1024→896	1024→768	1024→512
0.013	+.029 (.006)	+.059 (.012)	+.154 (.025)
0.038	+.158 (.016)	+.219 (.026)	+.480 (.029)
0.063	+.239 (.019)	+.323 (.028)	+.575 (.037)
0.088	+.282 (.026)	+.366 (.024)	+.622 (.033)
0.167	+.218 (.041)	+.348 (.033)	+.640 (.050)
0.300	+.291 (.047)	+.378 (.038)	+.714 (.045)
0.433	+.067 (.032)	+.253 (.041)	+.504 (.065)
0.567	+.029 (.023)	+.078 (.019)	+.360 (.055)
0.700	+.026 (.016)	+.032 (.023)	+.242 (.039)
0.833	−.027 (.020)	+.034 (.023)	+.218 (.053)
0.913	−.007 (.014)	+.039 (.013)	+.205 (.047)
0.938	−.039 (.026)	+.019 (.019)	+.200 (.049)
0.963	−.008 (.015)	+.026 (.016)	+.120 (.045)
0.988	+.018 (.028)	+.034 (.045)	+.217 (.052)
1.0 (endpoint)	+.034 (.011)	+.130 (.030)	+.406 (.049)

Table 8: **The floor, debiased** (scored in §4.4). **Endpoint**: at σ=1 the input is exactly ϵ on every grid, so the input-mediated data term vanishes by construction. **x-zero**: the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values $\widehat{\text{gap}}_\infty$ [bootstrap 95% CI over images] from nested draw sweeps (endpoint: N=12, $D = 4 \ldots 64$; x-zero: N=40, $D = 4 \ldots 32$). The two columns agree within CI at every route; our pre-registered falsification criterion (all endpoint gaps ≈ 0) is not met; and no member of the spectral family can produce a nonzero entry here.

route	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]	x-zero $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	1.005 [0.994, 1.016]	—
re-encoding control	−0.003 [−0.017, +0.008]	—
1024→896	+0.019 [+0.010, +0.030]	+0.034 [+0.017, +0.058]
1024→768	+0.056 [+0.043, +0.071]	+0.074 [+0.053, +0.094]
1024→512	+0.304 [+0.197, +0.424]	+0.283 [+0.232, +0.332]

Table 9: **Raw estimator** (historical record; superseded by Table 7): demotion gap by σ-bin, 1024-tier natives (N=40, bin-mean SEM ~0.02, re-encoding control $|\text{gap}_{\text{reenc}}| \leq 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. Raw values understate mid-σ gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); the debiased map corrects both.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap_{896} (ratio 0.875)	.110	.162	.148	.137	**.048**	**.030**	**.030**	.053
gap_{768} (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap_{512} (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
$\cos_{\text{floor}}$	.84	.65	.51	.65	.70	.77	.80	.83
$\|g\|$ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

Table 10: Iso-severity probe (**raw estimator**): gap by σ-bin on the 1280-tier cache (N=24, 4 draws/bin) against the corpus 1024→896 curve. The ratio-matched routes coincide within ~1.4 combined SEM at every bin despite a 1.6× difference in absolute target size; the same-native harsher-ratio route separates. *896's canonical 16-draw endpoint is −0.009 raw, +0.019 debiased (Table 8).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+.047	−.005	+.048	+.020	−.012
1280→1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280→1024 (ratio .80)	.170	.178	.111	.002	.061
1024→896 (ratio .875)	.132	.076	.077	.049	.100*

Table 11: Ratio-transfer run (**raw estimator**): 896-tier natives (N=40), pre-registered bar "gap_{768} within the re-encoding band at $\sigma \geq 0.5$": **fails** (residual 0.06–0.12, ~2× the 1024→896 plateau; ~half of it is expected estimator bias, §4.1; the separation is independently confirmed by the debiased same-grid floors, §4.4).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 12: Route-uniformity (in amplitude) of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_\epsilon\,[\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2; same-grid re-encoding control ≤ 0.02 everywhere). The three main routes coincide within $\sim \pm\, 0.02$ at every σ while their gradient floors span 0 to 0.3. This is a norm read; the direction-resolved follow-up (Appendix B) shows the underlying mismatch directions are image- and route-specific.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 13: Depth localization of the floor (**raw estimator**): mean per-block gap at σ=1, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry ~3× the late-block gap. The apparent content share (ep − xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adapter level (Table 8). Within a block every module type sits in 0.22–0.31 at $512/\sigma$=1, including the query and key rows of the fused QKV projection: landing-side uniformity, which is why the positional share needed an origin-side intervention to find.

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 14: Positional-interpolation intervention at the σ=1 endpoint (N=40, 16 draws; **raw** arm values; the paired Δ column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and ~30% of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024→896 (control)	−0.021 (.041)	−0.040 (.040)	—	—
1024→768	+0.080 (.048)	−0.001 (.039)	+0.081 (.031)	78%
1024→512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%